DANILLO XAVIER

Senior AI & Automation Engineer | Tech Lead

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8x QMS Contest LATAM 2025 Winner · 2x ICS Global Winner (2025) — People's Choice & Fellows · Information Security Manager (ISM) for Brazil Pipeline Delivery

PROFESSIONAL EXPERIENCE

May 2024 - Present Petrolina, Pernambuco, Brazil

BAYER CROP SCIENCE

TECHNOLOGY ENABLEMENT LEAD

Full-time

8x QMS LATAM 2025 · 2x ICS Global 2025 · ISM Brazil Pipeline Delivery

- Lead the development of AI, ML and automation solutions supporting corn and soybean R&D pipelines, integrating data platforms, decision-support systems and RPA to improve research productivity and operational efficiency.
- Coordinate cross-functional technical initiatives involving Python, BigQuery, Azure, Power Platform, Machine Learning and enterprise automation.

Technologies: Power BI · Power Automate · BigQuery · Azure · Python · ML · MCP · RAG · Agentic AI · React.js · JavaScript · SQL

Nov 2023 - Apr 2024 Belem do Sao Francisco, Pernambuco, Brazil · On-site

AGRODAN AGROPECUARIA RORIZ DANTAS

ADMINISTRATIVE MANAGER - SUPPLY CHAIN, CONTROLLERSHIP & HR OPERATIONS

Full-time

BRL 1.2M/year in labor reductions · BRL 500K/month in avoided requests · Inventory and people auditing

- Managed people and process indicators covering payroll, pay equity, time tracking, production, accounting, finance, procurement and supplier performance. Contribution: operational auditing and management of people controls, supporting BRL 1.2M/year in labor cost reductions and BRL 780K/year in absenteeism-related cost reductions.
- Audited physical and financial inventory movement, reorder point control, contract renegotiation and optimization through master invoices. Contribution: inventory control management and auditing, supporting reductions of BRL 1.8M/month in fast-moving inventory and BRL 150K/month in obsolete or slow-moving items.
- Automated request flows with AI-driven critical analysis, accounting class classification and real-time budget vs actual tracking with Telegram alerts. Contribution: auditing and management of request controls, associated with avoided expenditure of BRL 500K/month in unnecessary orders.
- Used SIA, the platform developed and evolved across my technology roles, to strengthen operational governance, auditing and management decision-making.

Technologies: React.js · Node.js · Oracle Database · Python · Power BI · Telegram Bot API · RPA · AI/ML · ERP Integration · SQL

Agrodan projects: SIA, data, mobile collection and business workflows

Mar 2021 - Nov 2023 Belem do Sao Francisco, Pernambuco, Brazil

AGRODAN AGROPECUARIA RORIZ DANTAS

HEAD OF TECHNOLOGY: IT AND INNOVATION

Full-time

85% of processes digitized · 150+ strategic views · 24 ETL jobs

- Held concurrently with the ICT Manager role from Mar 2021 to Nov 2023, combining technology and innovation leadership with ICT management.
- Led the company's digital transformation using PMBoK, Agile and gamification to institutionalize processes and an innovation culture.
- Identified opportunities, trained internal leaders and spread innovation mindset, methods and tools across the organization.
- Contribution: development and evolution of process and spraying audit tools supporting BRL 850K/year in pesticide savings, a 90% reduction in spraying/product-use issues and BRL 400K/year in fuel savings; tracked project impact and ROI for corporate decisions.
- Delivered digital forms, offline GPS-enabled collection, ERP integration, dashboards, ETLs and AI/ML automations, digitizing 85% of previously manual processes.
- Led the Agrodan Tech open innovation program, selecting seven startups; contract renegotiation supported BRL 800K/year in savings.

Technologies: React.js · Python · Power BI · ETL · Machine Learning · Deep Learning · IoT · ERP

Agrodan projects: SIA, data, mobile collection and business workflows

Oct 2019 - Nov 2023 Belem do Sao Francisco and Recife, Pernambuco · On-site

AGRODAN AGROPECUARIA RORIZ DANTAS

ICT MANAGER

Full-time

- From Mar 2021 to Nov 2023, held this role concurrently with Head of Technology: IT and Innovation.
- Managed IT staff routines, help desk operations, device estate, servers, backup and recovery, virtualization and high-availability initiatives.
- Administered Windows and Linux environments with Active Directory, Zabbix/GLPI, database synchronization, corporate networking, ERP customizations, data integration, dashboards, automations and IoT/security initiatives. Contribution: technical improvements to ERP modules and integrations reduced outsourcing needs, supporting BRL 240K/year in savings.

Technologies: Windows Server · Linux · Firewall · Active Directory · Zabbix · GLPI · Mikrotik · Unifi · Arduino · Raspberry Pi

Agrodan projects: SIA, data, mobile collection and business workflows

Jan 2019 - Oct 2019 Belem do Sao Francisco, Pernambuco, Brazil

AGRODAN AGROPECUARIA RORIZ DANTAS

IT BUSINESS PARTNER

Full-time

- Translated business needs into applied technologies with a strong focus on outcomes and transformation cadence.
- Led the creation and ongoing evolution of SIA, an integrated platform with real-time dashboards, tracking maps, offline synchronized collectors and AI-based modules for operational and strategic decisions.

Technologies: React.js · Node.js · SQL · Power BI · AI · Offline Sync

Agrodan projects: SIA, data, mobile collection and business workflows

PROFILE

Technology leader and AI & Automation Engineer with 10+ years of experience building enterprise software, ML solutions and intelligent automation for complex operational environments.
Currently Technology Enablement Lead at Bayer Crop Science, leading digital solutions for corn and soybean R&D pipelines. Experienced in Python, ML, cloud, data platforms and RPA, with a track record of delivering multi-million-BRL operational impact.
QMS LATAM 2025: Annual competition organized by Bayer LATAM's Operational Excellence team, evaluating projects that improve processes and quality and reduce costs.
ICS Global 2025: Bayer's annual internal congress that evaluates and recognizes the year's most innovative projects across Crop Science.

IMPACT HIGHLIGHTS

Results of initiatives developed and evolved between 2020 and April 2024, with contributions across different Agrodan roles. Monthly/annual bases retained, without an aggregated total.

- Saved BRL 850K/year in pesticide costs through process auditing tools.
  Spraying audit · PLANAGRO · Agrodan career context · Head of Technology · tools
- Reduced BRL 1.2M/year in labor through operational auditing and efficiency gains.
  Labor PDCA · Labor summary · Agrodan career context · Administrative management · controls
- Negotiated BRL 800K/year in contract savings and saved BRL 240K/year by reducing outsourcing needs in ERP modules.
  Agrodan career context · Head of Technology · tools · ICT management · ERP
- Reduced BRL 1.8M/month in fast-moving inventory costs and BRL 150K/month in obsolete or slow-moving inventory.
  Inventory management · Virtual inventory · Agrodan career context · Administrative management · controls
- Avoided BRL 500K/month in unnecessary requests with statistical and AI-driven auditing, plus BRL 400K/year in fuel savings.
  Purchase requests · Budget · Fleet refueling · Agrodan career context · Administrative management · controls · Head of Technology · tools
- Reduced 90% of spraying and product-use issues and cut BRL 780K/year in absenteeism-related costs.
  Spraying audit · Absenteeism · Agrodan career context · Head of Technology · tools · Administrative management · controls

EDUCATION

MAY 2026 — MAR 2027 (EXPECTED)

FM2S EDUCAÇÃO E CONSULTORIA

MBA OPERATIONAL EXCELLENCE — IN PROGRESS

- Business Administration, Management and Operations.
- Related independent Utilibox projects: VSM, SMED, PDSA, A3, Pareto and time study.

Related Lean and continuous improvement tools — Utilibox

2020 — 2025

CÉSAR SCHOOL

PROFESSIONAL DOCTORATE IN SOFTWARE ENGINEERING

- Completed in 2025.
- Thesis: Evolutionary Algorithms for Feature Selection in regression algorithms.
- Datasets: Irrigation (Kc), Mango sizes, Productivity

JUL 2022 — JUL 2023

UNIMAIS

POSTGRADUATE IN OCCUPATIONAL SAFETY ENGINEERING

- Capstone: AgriGuard — Georeferenced software for risk monitoring in agricultural environments

JAN 2017 — MAR 2018

UNINASSAU

POSTGRADUATE IN SOFTWARE ENGINEERING & PROJECT MANAGEMENT

- Capstone: Implementing SCRUM in a free web platform

2013 — 2016

UNIVASF

PRODUCTION ENGINEERING

- Capstone: F-AHP: Java software to apply the AHP multi-criteria decision method

ADDITIONAL EXPERIENCE

Feb 2012 - Oct 2019 Petrolina, Pernambuco, Brazil

D & E SOFTWARES E GAMES | SYRINX TECNOLOGIA | SILO STUDIO | CASA DOS ESMALTES

STARTUP FOUNDER & ENTREPRENEUR

Entrepreneurship

- Co-founded several startups and served as CEO in most of the initiatives.
- Built hands-on experience in market validation, product development, innovation and leadership under uncertainty.
- Independent projects were highlighted by Nokia in a curated list of Brazilian apps for smartphones.

Jan 2017 - Nov 2018 Petrolina Region, Brazil · On-site

GRUPO JD (LABRUNIER FARMS)

SENIOR SOFTWARE ENGINEER

Full-time

- Led the development of management and monitoring tools for grape packing operations.
- Built custom software, operational procedures and indicators that turned raw production data into actionable insights.
- Served as technical leader and co-creator of the Farm of Software initiative with advanced BI and interactive SQL-driven dashboards.

Sep 2016 - Dec 2016 Petrolina, Pernambuco, Brazil

GRUPO JD (LABRUNIER FARMS)

JUNIOR SOFTWARE ENGINEER

Full-time

- Implemented a production data collection system using RFID in the grape packing sector.
- Designed, monitored and managed operational indicators for both production lines and the IT team.
- Enabled real-time traceability, improved data accuracy and strengthened operational decision-making.

May 2015 - Aug 2015 Petrolina, Pernambuco, Brazil

HOSPITAL UNIVERSITARIO DE PETROLINA

INTERN - HOSPITAL PERFORMANCE & STRATEGIC PLANNING

Internship

- Developed a hospital performance indicator management system and supported the design and rollout of the 2015 Strategic Plan.
- Delivered training sessions, presentations and seminars to ensure effective adoption of the new tools.

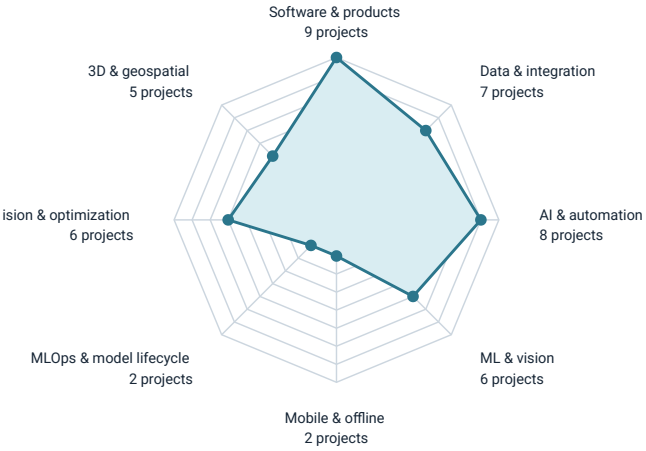
Internship project: hospital indicators and assessments in Java EE

Capabilities, stacks and application domains

Software engineering, AI, data and decisions connected to operations and product development.

21 projects in this overview **8 technical capabilities** **7 application domains**

Technical coverage · related projects



Stack · presence across projects

Python / data / ML	6
Web / APIs	15
SQL / persistence	11
Mobile / offline	2
3D / maps	4
Automation / integration	7

Projects using at least one technology in each family.

Python · SQL · JavaScript · TypeScript · Java
React · Node.js · Java EE · React Native
Oracle · PostgreSQL · SQLite · Supabase
scikit-learn · PyTorch · OpenCV · TensorFlow.js

Where I applied these capabilities

Agriculture & industry	3
Education & research	3
Finance & decisions	4
Operations & logistics	3
Healthcare	1
Digital products & CRM	5
Interaction & vision	4

Strengths connected to career experience

Leadership & operations
Bayer, Agrodan and Labrunier: technology, transformation, indicators and business integration.

Agentic AI / MLOps
Planejamento Xavier: multiagent coordination. TraderMind: model lifecycle. Globalcom: delivered architecture with RAG, MCP and audit, being implemented.

Research & continuous improvement
Doctoral EFS-AHP; Utilibox for decisions, Lean and finance; industrial vision at Agritech.NE.

Radar and bars: related project counts, with references below; these are not proficiency ratings. A project may support several capabilities. Backups and third-party references are excluded. Globalcom is presented separately as delivered architecture, being implemented.

Reference projects by capability

Software & products

SIA / SIAgro · EBSERH · The Game Project · ClassPlanner · Conversion CRM · Launch With Me · Planejamento Xavier · Utilibox · Personal finance

Data & integration

SIA / SIAgro · Agritech.NE · EFS-AHP · TraderMind · Cartola Oracle · Personal finance · Scrapers

AI & automation

Planejamento Xavier · ClassPlanner · Conversion CRM · DVR Person Alert · TraderMind · mocap · instabot · Scrapers

ML & vision

EFS-AHP · TraderMind · DVR Person Alert · mocap · Genetic algorithms · Agritech.NE

Mobile & offline

SIA / SIAgro · Planejamento Xavier

MLOps & model lifecycle

TraderMind · DVR Person Alert

Decision & optimization

Utilibox · EFS-AHP · Genetic algorithms · Cartola Oracle · Logistics · SIA / SIAgro

3D & geospatial

Dente de Leão · mocap · Movimento 3D · Logistics · SIA / SIAgro

CORE STACK

Python · SQL · JavaScript · TypeScript · Java · React · Node.js · React Native / Expo · Oracle / PostgreSQL · BigQuery · Azure · Power Platform · scikit-learn · YOLO / OpenCV · LLM APIs · ETL

SKILLS

AI, agents and automation

LLM integration, structured outputs, context, fallback and rule-based coordination; computer vision and process automation.

Planejamento Xavier, ClassPlanner, CRM, DVR and professional experience at Bayer.

Machine Learning and research

Feature selection, evolutionary algorithms, comparative experiments, ensembles and temporal validation.

EFS-AHP doctoral research, TraderMind and exploratory irrigation modeling.

Software engineering

Web and mobile applications, domain modeling, APIs, persistence, offline collection, synchronization and desktop integrations.

SIA/SIAgro, EBSERH, The Game Project and independent products.

Data and decisions

ETL, ERP integration, SQL, indicators, operational auditing, multicriteria methods and constrained optimization.

Agrodan, Labrunier, Utilibox, Cartola Oracle and Logística.

Leadership and operations

Technology management, business alignment, team development, PMBOK, Scrum, Kanban, governance, infrastructure and security.

Technology Enablement Lead, Head of Technology, IT Manager and IT Business Partner roles.

Products and applied methods

Product development, multicriteria methods, operations research, financial analysis, continuous improvement, learning and knowledge management.

Utilibox: AHP, TOPSIS, PROMETHEE, queues, VSM/SMED, PDSA, NPV/IRR, Narra and UtiliNotas; SIA: quality, packing and operations.Related education: ongoing FM2S MBA in Operational Excellence.

Agentic AI architecture and governance

Design of orchestrators, specialist agents, approved tools, contextual RAG, independent auditing and human approval.

Globalcom Certification OS — Modern Concept delivered architecture; implementation in progress.

MLOps and model lifecycle

Data and feature pipelines, checkpoints, model artifacts, temporal validation, API serving, supervision, retraining and observability.

TraderMind — quantitative research and demonstration operation integrated with MetaTrader5.

Algorithms and problem solving

Data structure practice, algorithmic strategies, complexity analysis and edge-case handling.

120 hours of LeetCode practice.

Computer vision and industrial integration

Platform development with fruit counting/tracking, ERP/sorting-machine

TOOLS

Languages: Python, SQL, JavaScript, TypeScript and Java. · **Web, APIs and products:** React, Next.js, Astro, Node.js, Express, Flask, Java EE, JSF, PrimeFaces and Supabase. · **Mobile and offline:** React Native, Expo, AsyncStorage, IndexedDB and Dexie. · **Data and persistence:** Oracle, PostgreSQL, SQL Server, SQLite, BigQuery, pandas, NumPy, Parquet and Sequelize. · **ML and computer vision:** scikit-learn, XGBoost, LightGBM, CatBoost, SciPy, DEAP, PyTorch, YOLO, OpenCV, TensorFlow.js and MediaPipe. · **Decision and visualization:** AHP, TOPSIS, PROMETHEE, OR-Tools, GLPK, Power BI, Leaflet, Three.js and React Three Fiber. · **Automation and integration:** LLM APIs, Gemini, Playwright, Puppeteer, Telegram Bot API, Power Platform and RPA. · **Cloud and operations:** Azure, GCP, AWS, Docker, Kubernetes, Terraform, Airflow, dbt, MLflow, Windows/Linux, Active Directory and Zabbix/GLPI. · **Architecture-specified stack — Globalcom Agentic AI:** React/TypeScript, Node.js, Python/FastAPI, Supabase/PostgreSQL, pgvector, RAG, MCP and Playwright; NestJS/Fastify and BullMQ as suggested alternatives, optional Temporal. · **MLOps — TraderMind:** Python, pandas, NumPy, Parquet, scikit-learn, DEAP/EFS-AHP, joblib, Flask, MetaTrader5; checkpoints, temporal validation, supervisor, logs and dashboard. · **Packing AI integration:** Computer vision, Deep Learning, cameras, APIs, ERP, sorting machine, dashboard, Telegram and a dedicated server; CRISP-DM project methodology.

integration, weight estimation, dashboard and operational alerts.

Agrodan / Agritech.NE — applied R&D project, March 2024 final report.

Projects and applied engineering at Agrodan

SIA / SIAgro ecosystem · professional experience

Computer vision platform development for packing operations

Worked as a developer on the platform project during my Agrodan employment period, connecting fruit counting and tracking to operational packing monitoring.

- Platform for identifying and recording mangoes on the reject conveyor, with counting and tracking to avoid duplicates.
- Connected the counting application, industrial data API, dashboard and notifications; the report describes two operating cameras.

Engineering and technologies: Computer vision · Deep Learning · Cameras · APIs · ERP · Sorting-machine integration · Dashboard · Telegram · Dedicated server · CRISP-DM.

Agrodan career context · Technical details

ERP, sorting-machine integration and batch traceability

The platform connects incoming batches, packed volume and rejected fruit to track fruit origin and support batch transitions.

- API access to sorting-machine and packing data, combined with ERP batch records and variety, harvest and handling information.
- Fruit counts are combined with the sorting-machine size distribution and variety-specific weight ranges to estimate reject weight.

Engineering and technologies: Computer vision · Deep Learning · Cameras · APIs · ERP · Sorting-machine integration · Dashboard · Telegram · Dedicated server · CRISP-DM.

Agrodan career context · Technical details

Dashboard, alerts and operational decision support

Developed the monitoring and notification platform to provide operators with processing information and batch-change timing.

- Monitored integrated production data and prepared Telegram alerts when processed volume approaches incoming volume.
- The scope includes dedicated-server counting and monitoring applications; the report records training and follow-up in February 2024.

Engineering and technologies: Computer vision · Deep Learning · Cameras · APIs · ERP · Sorting-machine integration · Dashboard · Telegram · Dedicated server · CRISP-DM.

Agrodan career context · Technical details

Project outcomes and industrial process improvement

Platform integration into the packing workflow produced project-level outcomes recorded in the final report, covering throughput, downtime and operating costs.

- Reported throughput increased from 30 to 37.5 boxes per hour; daily control-change downtime decreased from 45 to 5 minutes.
- The report records a 17% reduction in operating costs associated with labor and processing time.

Engineering and technologies: Computer vision · Deep Learning · Cameras · APIs · ERP · Sorting-machine integration · Dashboard · Telegram · Dedicated server · CRISP-DM.

Agrodan career context · Technical details

PLANAGRO — inputs, applications and agronomic rules

Developed agricultural planning components in PLANAGRO, translating input application, dosage, withholding periods, drift and budgets into rules and interfaces connected to operational data.

- Domain rules represent dosage, withholding periods, drift and application constraints.
- Planning interfaces connect operational inputs and budget information to the application workflow.

Engineering and technologies: React · JavaScript · Node.js · operational APIs · agronomic rules.

Agrodan career context · Technical details

Irrigation — scheduling, genetic algorithms and Kc modeling

Explored irrigation scheduling, genetic algorithms and neural modeling of crop coefficient Kc, working with variable encoding, normalization, network parameters and operational backend integration.

- Categorical encoding and normalization prepare variables such as farm, phase, month and soil.

- Neural training, serialized model restoration and evolutionary experiments explore irrigation-related decisions.

Engineering and technologies: JavaScript · Node.js · genetic algorithms · neural network · normalization.

Exploratory modeling; out-of-sample performance is not established.

Agrodan career context · Technical details

Spraying audit — maps and field execution

Built spraying-audit components using spatial and operational data, relating trajectories, speed, field blocks and services to support investigation of field execution.

- Maps connect movement and speed to field blocks and operational records.
- Audit interfaces make execution patterns available for operational review.

Engineering and technologies: React · geospatial data · trajectory · speed · operational APIs.

Agrodan career context · Technical details

Dry matter — harvest forecasting and readiness

Implemented dry-matter analysis and harvest-forecast components connecting readings, varieties and parameters to charts and monitoring rules.

- Equations and parameters relate dry-matter observations to harvest readiness.
- Charts and operational interfaces support monitoring by variety and field context.

Engineering and technologies: React · JavaScript · curves and equations · charts · field data.

Agrodan career context · Technical details

Field notebook and purchasing — local data and synchronization

Developed field-data collection and purchase-request workflows with local storage, process components and synchronization routines integrated with domain-specific backend services.

- Dexie/IndexedDB stores field and purchasing data locally.
- Synchronization handlers and domain APIs connect local records to the backend.

Engineering and technologies: React · Dexie · IndexedDB · Node.js · synchronization.

Agrodan career context · Technical details

UGB management, people and improvement suggestions

Modeled business-unit and people-management applications and improvement-suggestion workflows combining organizational records, history, assessments, roles and documents.

- Business units, employees and roles share domain models and reusable components.
- Assessments, history, 5S and improvement workflows organize management information.

Engineering and technologies: React · Node.js · organizational modeling · assessments · history.

Agrodan career context · Technical details

Commercial monitoring, forecasts and pallet movements

Developed commercial and logistics components connecting records and forecasts to management interfaces and operational pallet/container data.

- Commercial entries, forecasts and exchange-rate information support operational views.
- Pallet and container movements link logistics records to management interfaces.

Engineering and technologies: React · Node.js · commercial data · pallets · containers.

Agrodan career context · Technical details

Modular ETL and operational data quality

Structured integration pipelines with reusable importers, batch processing and domain rules for production, packing, quality and people, organizing operational data for analysis and decisions.

- Node.js and SQL importers separate extraction, transformation and domain-specific processing.
- Batch persistence, joins and aggregation connect ERP and operational records to indicators.

Engineering and technologies: Node.js · SQL · Oracle · Sequelize · ETL · batch import.

Agrodan career context · Technical details

SIAgro — mobile operational data collection

Developed SIAgro mobile modules for work orders, production records and quality using React Native/Expo, business-object modeling, local persistence and operational indicators.

- Mobile workflows organize work orders, employees, targets and quality records.
- Reusable local persistence and API integration connect collection and dashboard components.

Engineering and technologies: React Native · Expo · AsyncStorage · React Navigation · APIs.

Agrodan career context · Technical details

IT operations management — Agrodan

Managed IT operations, integrating support, servers, networks, continuity, ERP and IoT initiatives with delivery of technology services and solutions.

- Professional responsibilities cover support, Windows/Linux administration, Active Directory and corporate networks.
- Backup/recovery, virtualization, monitoring, security and IoT belong to the declared IT management scope.

Engineering and technologies: Windows Server · Linux · Active Directory · Zabbix · GLPI · networks · IoT.

Agrodan career context · Technical details

QMS — audits, records and quality management

Developed quality-management modules organizing audits, records and roles with process follow-up and local storage.

- Audit, configuration and responsibility models support management interfaces.
- Dexie/IndexedDB local data and synchronization handlers connect workflows.

Engineering and technologies: React · JavaScript · Dexie · IndexedDB · synchronization.

Agrodan career context · Module

Field quality — checklists and operational assessments

Developed field-quality assessment workflows with checklists, responses and integration between interfaces and the domain backend.

- Checklists and responses use Dexie/IndexedDB local persistence.
- Synchronization routines and corresponding backend routes were located in the archive.

Engineering and technologies: React · JavaScript · Dexie · IndexedDB · Node.js · domain APIs.

Agrodan career context · Module

Packing and box handling — operational interfaces

Developed packing and box-handling interfaces organizing access to operations and connecting events to monitoring components.

- Separate packing and box-handling components structure the corporate frontend.
- Socket event integration is present in the module code.

Engineering and technologies: React · JavaScript · socket events · operational interfaces.

Agrodan career context · Module

Utilibox — products and applied methods

Decisions, operations research, production engineering, finance and knowledge · independent product

AHP Online — multicriteria decisions and consistency

Developed a multicriteria decision tool organizing criteria, comparisons and alternatives into an analysis and project-export workflow.

- Saaty-scale comparisons and consistency calculations assess the coherence of judgments.
- Interactive decision modeling presents results and supports project export.

Engineering and technologies: Astro · JavaScript · AHP · comparison matrices · consistency.

Open tool — ahp-online

TOPSIS and PROMETHEE — rankings and sensitivity

Implemented tools for comparing alternatives with configurable criteria, weights and preferences, applying multicriteria methods to management decisions.

- TOPSIS ranks alternatives by closeness to the ideal solution, with configurable weights and criterion directions.
- PROMETHEE I and II include preference functions and real-time sensitivity analysis.

Engineering and technologies: Astro · JavaScript · TOPSIS · PROMETHEE · sensitivity analysis.

Open tool — topsis · Open tool — promethee

Queueing theory — capacity, waiting and service sizing

Developed calculators and visualizations relating demand, service capacity and waiting, supporting study and comparison of service configurations.

- Calculation routines implement M/M/1, M/M/c, M/D/1, M/M/1/K and Erlang-B/C models.
- Server-count comparisons and sensitivity curves explore changes in arrival rates.

Engineering and technologies: JavaScript · operations research · queueing models · sensitivity curves.

Open tool — teoria-das-filas · Open tool — queueing-theory

VSM, SMED, PDSA and A3 — production engineering and continuous improvement

Translated process-improvement methods into digital tools for mapping flows, analyzing setup, structuring problems and tracking learning cycles.

- VSM models steps, processing, waiting, uptime, inventory and bottlenecks, with lead-time, takt and PCE metrics.
- SMED explores internal/external activities, conversion and parallelization; PDSA and A3 structure follow-up and action plans.

Engineering and technologies: Astro · JavaScript · Lean · VSM · SMED · PDSA · A3.

MBA Operational Excellence · Open tool — vsm · Open tool — smed · Open tool — pdsa-manager · Open tool — folha-a3

Economic analysis — NPV, IRR, payback and scenarios

Implemented economic-analysis tools transforming cash flows, costs and financing conditions into comparable indicators.

- Simple and discounted payback, NPV and IRR calculations support investment analysis.
- Complementary tools address WACC, break-even and Price, SAC and SAM repayment systems.

Engineering and technologies: JavaScript · financial mathematics · cash flows · investment analysis.

Open tool — calculadora-payback-vpl-tir · Open tool — calculadora-wacc · Open tool — simulador-ponto-de-equilibrio · Open tool — simulador-emprestimo

Pareto, GUT and SWOT — prioritization and management analysis

Developed management tools organizing problems, criteria and scenarios into rankings, charts and decision structures.

- Pareto includes cumulative curves; GUT ranks severity, urgency and trend; SWOT supports cross-analysis of strategies.
- Decision trees include chance/outcome nodes, expected monetary value and optimal-path identification.

Engineering and technologies: JavaScript · Pareto · GUT · SWOT · EMV · decision visualization.

MBA Operational Excellence · Open tool — analise-pareto · Open tool — matriz-gut · Open tool — matriz-swot · Open tool — arvore-decisao

Narra and UtiliNotas — knowledge, context and persistence

Built narrative and knowledge-organization applications with related entities, linked notes and contextual assistance.

- Markdown editing, internal links and file trees connect note organization to usage context.
- IndexedDB persistence manages transactions and failures; AI assistance resides in dedicated modules.

Engineering and technologies: Astro · JavaScript · Markdown · IndexedDB · entity modeling · contextual AI.

Open tool — narra · Open tool — utilinotas

Flashcards — spaced repetition and study tracking

Implemented a study application organizing cards and reviews through spaced repetition, retention tracking and export.

- SM-2, due-card selection, suspension/resumption and review forecasting organize study.
- Anki Desktop TSV and PDF exports support structured decks and content.

Engineering and technologies: JavaScript · SM-2 · local persistence · TSV · PDF.

Open tool — anki-flashcards

Applied research and education

Doctoral research, hospital internship and university projects

EFS-AHP — doctoral research framework

Developed EFS-AHP as doctoral research combining feature selection, evolutionary algorithms and multicriteria decision analysis, comparing ML alternatives and analyzing performance, dimensionality and computational cost.

- Experiments compare methods including Random Forest, LASSO and dimensionality-reduction approaches.
- Python notebooks and evolutionary routines organize datasets, baselines and statistical analysis; AHP supports comparison of alternatives.

Engineering and technologies: Python · scikit-learn · pandas · NumPy · SciPy · DEAP · AHP.

Doctoral research · Public repository — EFS-AHP

Hospital indicators and academic gamification

During the hospital internship, developed a Java EE web system for indicators and assessments. At university, built a project-management application with gamification and persistence, exploring progression rules and task organization.

- EBSERH: Java EE, EJB, JSF, PrimeFaces and JPA/Hibernate for hospital indicators and assessments.
- The Game Project: persistent tasks, progression rules, scheduling and an interactive project timeline.

Engineering and technologies: Java EE · EJB · JSF · PrimeFaces · JPA/Hibernate · JPQL · Quartz.

Hospital internship · Context and projects · Public repository — EBSERH · Public repository — The Game Project

Agentic AI — Globalcom Certification OS

Modern Concept · delivered educational certification and compliance architecture · implementation in progress

Globalcom Certification OS

Designed and delivered the functional and technical architecture for Globalcom Certification OS. The scope integrates 10 specialist agents, an orchestrator and independent auditor, RAG, persistent workflows, domain modeling, multi-tenant security and API/MCP/RPA integrations. The delivered architecture guides the ongoing implementation.

Specified stack: TypeScript · React · Node.js · Python · FastAPI · Supabase · PostgreSQL · pgvector · RAG · MCP · Playwright · RLS · RBAC · Queues and workers.

Architecture delivery completed; implementation in progress. Operational results have not yet been declared.

- Agentic AI architecture and specialist orchestration
- Managed RAG and document intelligence
- Workflow, events and persistent execution
- AI governance, security and continuous audit
- API, MCP and RPA integrations for external operations
- Certification domain and product modeling
- Consulting, solution architecture and roadmap

MLOps — TraderMind

ML pipeline, validation, inference, supervision and maintenance · research and demo operation

MetaTrader5 desktop connector — TraderMind

Built a Python connector to the MetaTrader5 desktop terminal, integrating market/account data with the documented research and execution workflow.

- The connector uses the MetaTrader5 package and requires an authenticated local terminal.
- Candles and account data connect policy research with separate API, dashboard and execution components.

Engineering and technologies: Python · MetaTrader5 · Flask · SQLite · local APIs.

Context and projects · Technical details

Data pipelines, features and quality contracts

Integrated extraction, data updates, dataset construction and feature selection into operational TraderMind pipeline stages.

- Market data and multitimeframe indicators, Parquet datasets and evolutionary feature selection with EFS-AHP.
- Stage checks cover temporal integrity, feature availability and health, configuration consistency and artifact synchronization.

Engineering and technologies: Python · pandas · NumPy · Parquet · DEAP · EFS-AHP.

Context and projects · Technical details

Training, checkpoints and artifact management

Organized experiments, training and execution recovery while preserving the relationship between models, features and parameters.

- Classifier training and comparison routines, feature selection and entry and early-exit models.
- Checkpoints resume evolutionary search; serialized models, feature lists, configuration and logs support workflow reproduction.

Engineering and technologies: Python · scikit-learn · XGBoost · LightGBM · CatBoost · joblib · checkpoints.

Context and projects · Technical details

Temporal validation, leakage and out-of-sample evaluation

Separated selection, calibration and temporal evaluation to compare models and policies without contaminating the final evaluation period.

- Walk-forward, temporal holdouts, backtests and leakage audits complement training; causal external-data alignment avoids future information.
- Policies are selected before final OOS testing; comparisons between logged operation and replay/backtest help investigate discrepancies.

Engineering and technologies: Python · temporal validation · walk-forward · OOS · backtesting.

Context and projects · Technical details

Model serving and MetaTrader5 integration

Connected ML artifacts, an inference service and the desktop terminal to bridge research and demonstration operation.

- A Flask API loads models and their features by pair/direction, returns predictions and probabilities and supports exit checks.
- A Python connector integrates data and execution with MetaTrader5; service endpoints expose health, status, loaded models and uptime.

Engineering and technologies: Python · Flask · REST APIs · joblib · MetaTrader5.

Context and projects · Technical details

Orchestration, retraining and operational updates

Coordinated data, training, evaluation and serving cycles through distinct execution modes.

- The supervisor organizes extraction, datasets, EFS-AHP, training, backtesting and serving; it distinguishes data updates without retraining from full retraining cycles.
- Runs use identifiers and logs; scheduling and operational checks support pipeline maintenance.

Engineering and technologies: Python · supervisor · scheduling · logs · pipeline automation.

Context and projects · Technical details

Observability and model follow-up

Connected the operational dashboard, pipeline stage health and performance review using recent data.

- Dashboard and health records track stages, training and operation; APIs expose model and service availability.
- OOS window updates and comparisons between logged trades and backtests support regime-change analysis and model review decisions.

Engineering and technologies: Python · Flask · dashboard · logs · OOS · monitoring.

Context and projects · Technical details

AI, multiagent systems and Machine Learning projects

Independent development and quantitative research

Six-agent planning council — Planejamento Xavier

Implemented a council of six specialized agents to support planning and replanning, with evidence for each recommendation, deterministic conflict resolution and traceable actions.

- Strategy, execution, risk, learning, integrity and mentoring agents analyze the same planning snapshot.
- Claims carry evidence, confidence, polarity, weight and node references; priorities and deterministic tie-breaking resolve conflicts.

Engineering and technologies: TypeScript · Deno · Supabase · typed contracts · conflict resolution.

Rule-based agent coordination; LLM integration in a separate layer.

Context and projects · Live website — Planejamento Xavier

Financial signal ensemble — TraderMind

Designed multiagent policy experiments with signal voting, quorum and simulated annealing, separating calibration, temporal validation and out-of-sample evaluation.

- Twelve signals support ternary voting, minimum active-agent counts and agreement thresholds.
- Policies choose whether to follow, invert or ignore signals; the selected policy is frozen before final out-of-sample evaluation. Backward as-of joins respect publication delays.

Engineering and technologies: Python · pandas · NumPy · simulated annealing · temporal validation.

Signal-agent research; financial returns were not validated in this review.

Context and projects · Technical details

Shared AI integration engine — Planejamento Xavier

Developed a shared AI integration layer with provider adapters, structured responses, validation, time budgets and task-specific failure handling.

- Provider capabilities include JSON, text and grounded search; adapters support Gemini, DeepSeek, OpenAI and compatible APIs.
- Global and per-attempt time budgets, output rejection, fallback and provider health tracking organize execution. Grounded search preserves source references.

Engineering and technologies: TypeScript · Deno · LLM APIs · JSON · telemetry · fallback.

Context and projects · Live website — Planejamento Xavier

AI-assisted plan review and operation contracts

Modeled AI-assisted review proposals with reference validation, alternatives, operations and follow-up, connecting suggested changes to the actual plan.

- Authorized context contains real IDs, capacity, diagnostics and prior decisions.
- Normalized operations include dependencies, affected occurrences, expected update timestamps, context fingerprints and validation issues.

Engineering and technologies: TypeScript · Supabase · operation contracts · reference validation.

Context and projects · Live website — Planejamento Xavier

AI-assisted lesson planning — ClassPlanner

Integrated Gemini lesson-plan generation into a web application using pedagogical context, Brazilian BNCC curriculum references and structured objectives, methodology and assessment.

- Prompts combine class, subject, duration, resources, assessment and curriculum skills.
- Structured JSON responses are parsed into the planning workflow.

Engineering and technologies: React · Node.js · Express · SQLite · Gemini · BNCC.

Context and projects · Technical details

Hybrid conversational CRM — Instagram Conversion CRM

Built a hybrid AI-and-rules conversation flow with interaction history, action contracts, failure handling and human handoff.

- Conversation state and history provide context for the decision engine.
- Action normalization, retries and handoff connect suggestions to the CRM workflow.

Engineering and technologies: Node.js · Express · SQLite · Gemini · conversation history.

Context and projects · Technical details

TraderMind — experiments and inference integration

Structured quantitative experiments and inference components, connecting data preparation and model/policy evaluation to an API and the MetaTrader5 terminal.

- Time-series preparation and feature engineering feed model and policy experiments.
- API and desktop connector components keep research, inference and execution responsibilities separate.

Engineering and technologies: Python · scikit-learn · XGBoost · pandas · Flask · MetaTrader5.

Context and projects · Technical details

DVR — feedback, datasets and classifier training

Implemented a computer-vision workflow with detection, example review and custom classifier training from a pretrained model, integrating epoch monitoring, checkpoints, temporal stabilization and persistent alerts.

- Reviewed examples form a training dataset; transfer learning and callbacks organize training and checkpoints.
- Temporal voting and a persistent alert outbox connect detection to operational notifications.

Engineering and technologies: Python · OpenCV · Ultralytics/YOLO · PyTorch · Flask · RTSP.

Context and projects · Technical details

Mocap — browser inference and visualization

Developed browser pose-estimation prototypes integrating inference, keypoint visualization and temporal smoothing to explore motion capture and 3D interaction.

- Existing pose models process video and webcam input in the browser.
- Keypoint rendering, smoothing and 3D resources connect inference to interactive visualization.

Engineering and technologies: JavaScript · TensorFlow.js · MoveNet · BlazePose · MediaPipe · WebGL.

Interaction and visualization prototypes and studies.

Context and projects · Technical details

Optimization and systems integration

Quantitative research and independent development

Cartola and logistics — decisions under constraints

Developed pipelines connecting data and optimization to decisions, exploring team selection under budget/formation constraints and routing with geographical visualization.

- Cartola combines performance data and probabilistic estimates with constrained team selection.
- Logistics prototypes combine routes, maps and optimization experiments.

Engineering and technologies: Python · pandas · Parquet · OR-Tools/CBC · React · Leaflet.

Context and projects · Public repository — Cartola Oracle

Industrial R&D experience — Agrodan / Agritech.NE

Fruit counting and traceability platform developer

During the Agrodan employment period · final report: March 2024

Worked as a platform developer during the Agrodan employment period, connecting fruit counting and tracking on the reject conveyor with ERP and sorting-machine data, an operational dashboard and Telegram alerts for batch traceability and changeover decisions. The final project report is dated March 2024.

Project and technical contribution

Additional experience and technical practice

LeetCode — 120 hours of algorithm and data structure practice

Complementary software engineering problem-solving practice, including time and space complexity, data structure selection and edge-case debugging.

Topics and reference challenges

- **Arrays, strings and hashing:** Two Sum, Group Anagrams, Product of Array Except Self.
- **Two pointers, sliding windows and binary search:** Longest Substring Without Repeating Characters, Binary Search, Search in Rotated Sorted Array.
- **Stacks, linked lists and intervals:** Valid Parentheses, Reverse Linked List, Merge Intervals.
- **Trees, graphs and heaps:** Binary Tree Level Order Traversal, Number of Islands, Course Schedule, Top K Frequent Elements.
- **Dynamic programming and backtracking:** Coin Change, House Robber, Longest Increasing Subsequence, Subsets.

Project details · applications, methods and technologies

Projects, applications and technologies

1. Projects and career context
2. Scope of professional and technical experience
3. Technical cases — multiagent coordination
4. Technical cases — AI integration
5. Technical cases — Machine Learning and research
6. Technical cases — computer vision
7. Technical cases — agricultural and business applications
8. Technical cases — data engineering
9. Technical cases — mobile and enterprise web
10. Technical cases — decision support
11. Technical cases — products and knowledge
12. Technical cases — infrastructure and desktop integration
13. Technical cases — 3D and spatial visualization
14. Technical cases — Globalcom Agentic AI solution architecture
15. Technical cases — TraderMind MLOps and model lifecycle
16. Technical cases — Agrodan industrial computer vision platform
17. Doctoral, hospital internship and university projects
18. SIA — business modules
19. SIA / SIAgro — data and mobile workflows
20. Complete project and reference catalog
21. Utilibox — applications and product domains
22. Capabilities and their project origins
23. Project and career context

Projects and career context

Agrodan — professional SIA / SIAgro ecosystem

Professional tenure: Jan 2019–Apr 2024. Head of Technology and ICT Manager were concurrent roles from Mar 2021 to Nov 2023. Field operations, packing, people, planning, purchasing, quality and commercial modules are associated with this tenure. Results of initiatives developed and evolved between 2020 and April 2024, with contributions across different Agrodan roles. Role periods identify tenure in each position. Indicators retain their monthly or annual basis and are not added into a total savings figure.

- IT Business Partner
- ICT Manager
- Head of Technology
- Administrative management — supply chain, controlling and HR operations

Professional experience · Impact results

EFS-AHP — doctoral research

Feature selection, evolutionary algorithms and multicriteria analysis are associated with doctoral research at CÉSAR School.

Education

EBSERH and The Game Project

EBSERH belongs to the hospital internship. The Game Project was university work. Other projects include independent products, research, studies and entrepreneurial initiatives; individual company/date attribution is not inferred.

Scope of professional and technical experience

The trajectory combines software, data and automation engineering with technology leadership and applications in agriculture, operations, education, finance, production, planning and digital products.

Depth is described by implemented domain models, persistence, integrations, data pipelines, research methods, interfaces and prototypes. The

number of modules does not establish independent products or a seniority rating.

Technical cases — multiagent coordination

Six-agent planning council — Planejamento Xavier

Independent projects, prototypes and research

Implemented a council of six specialized agents to support planning and replanning, with evidence for each recommendation, deterministic conflict resolution and traceable actions.

Application area: Personal planning, productivity and professional development.

Users and context: People managing goals, activities and development plans.

Objective: Connect goals, capacity and execution through AI-assisted planning and review.

Engineering and implementation

- Strategy, execution, risk, learning, integrity and mentoring agents analyze the same planning snapshot.
- Claims carry evidence, confidence, polarity, weight and node references; priorities and deterministic tie-breaking resolve conflicts.

Technologies and methods: TypeScript · Deno · Supabase · typed contracts · conflict resolution

Project links

Live website — Planejamento Xavier

Financial signal ensemble — TraderMind

Independent projects, prototypes and research

Designed multiagent policy experiments with signal voting, quorum and simulated annealing, separating calibration, temporal validation and out-of-sample evaluation.

Application area: AI, Machine Learning and MLOps for quantitative research.

Users and context: Financial-market model and strategy research and experimentation.

Objective: Structure the data, training, temporal validation, inference and model supervision lifecycle; integrate signals and policies with MetaTrader5.

Engineering and implementation

- Twelve signals support ternary voting, minimum active-agent counts and agreement thresholds.
- Policies choose whether to follow, invert or ignore signals; the selected policy is frozen before final out-of-sample evaluation. Backward as-of joins respect publication delays.

Technologies and methods: Python · pandas · NumPy · simulated annealing · temporal validation

Technical cases — AI integration

Shared AI integration engine — Planejamento Xavier

Independent projects, prototypes and research

Developed a shared AI integration layer with provider adapters, structured responses, validation, time budgets and task-specific failure handling.

Application area: Personal planning, productivity and professional development.

Users and context: People managing goals, activities and development plans.

Objective: Connect goals, capacity and execution through AI-assisted planning and review.

Engineering and implementation

- Provider capabilities include JSON, text and grounded search; adapters support Gemini, DeepSeek, OpenAI and compatible APIs.
- Global and per-attempt time budgets, output rejection, fallback and provider health tracking organize execution. Grounded search preserves source references.

Technologies and methods: TypeScript · Deno · LLM APIs · JSON · telemetry · fallback

Project links

Live website — Planejamento Xavier

AI-assisted plan review and operation contracts

Independent projects, prototypes and research

Modeled AI-assisted review proposals with reference validation, alternatives, operations and follow-up, connecting suggested changes to the actual plan.

Application area: Personal planning, productivity and professional development.

Users and context: People managing goals, activities and development plans.

Objective: Connect goals, capacity and execution through AI-assisted planning and review.

Engineering and implementation

- Authorized context contains real IDs, capacity, diagnostics and prior decisions.
- Normalized operations include dependencies, affected occurrences, expected update timestamps, context fingerprints and validation issues.

Technologies and methods: TypeScript · Supabase · operation contracts · reference validation

Project links

Live website — Planejamento Xavier

AI-assisted lesson planning — ClassPlanner

Independent projects, prototypes and research

Integrated Gemini lesson-plan generation into a web application using pedagogical context, Brazilian BNCC curriculum references and structured objectives, methodology and assessment.

Application area: Education and lesson planning.

Users and context: Teachers.

Objective: Organize lesson plans using curriculum context, objectives, methodology and assessment.

Engineering and implementation

- Prompts combine class, subject, duration, resources, assessment and curriculum skills.
- Structured JSON responses are parsed into the planning workflow.

Technologies and methods: React · Node.js · Express · SQLite · Gemini · BNCC

Hybrid conversational CRM — Instagram Conversion CRM

Independent projects, prototypes and research

Built a hybrid AI-and-rules conversation flow with interaction history, action contracts, failure handling and human handoff.

Application area: Marketing, sales and customer relationships.

Users and context: Conversational sales and customer service operations.

Objective: Organize the sales funnel and support conversational decisions with history and suggested actions.

Engineering and implementation

- Conversation state and history provide context for the decision engine.
- Action normalization, retries and handoff connect suggestions to the CRM workflow.

Technologies and methods: Node.js · Express · SQLite · Gemini · conversation history

Technical cases — Machine Learning and research

EFS-AHP — doctoral research framework

Doctoral research

Developed EFS-AHP as doctoral research combining feature selection, evolutionary algorithms and multicriteria decision analysis, comparing ML alternatives and analyzing performance, dimensionality and computational cost.

Application area: Data science, Machine Learning and multicriteria decision making.

Users and context: Applied doctoral research in Software Engineering.

Objective: Select features and compare model alternatives by performance, dimensionality and cost.

Engineering and implementation

- Experiments compare methods including Random Forest, LASSO and dimensionality-reduction approaches.
- Python notebooks and evolutionary routines organize datasets, baselines and statistical analysis; AHP supports comparison of alternatives.

Technologies and methods: Python · scikit-learn · pandas · NumPy · SciPy · DEAP · AHP

Project links

Public repository — EFS-AHP

TraderMind — experiments and inference integration

Independent projects, prototypes and research

Structured quantitative experiments and inference components, connecting data preparation and model/policy evaluation to an API and the MetaTrader5 terminal.

Application area: AI, Machine Learning and MLOps for quantitative research.

Users and context: Financial-market model and strategy research and experimentation.

Objective: Structure the data, training, temporal validation, inference and model supervision lifecycle; integrate signals and policies with MetaTrader5.

Engineering and implementation

- Time-series preparation and feature engineering feed model and policy experiments.
- API and desktop connector components keep research, inference and execution responsibilities separate.

Technologies and methods: Python · scikit-learn · XGBoost · pandas · Flask · MetaTrader5

Technical cases — computer vision

DVR — feedback, datasets and classifier training

Independent projects, prototypes and research

Implemented a computer-vision workflow with detection, example review and custom classifier training from a pretrained model, integrating epoch monitoring, checkpoints, temporal stabilization and persistent alerts.

Application area: Security, video monitoring and computer vision.

Users and context: Camera monitoring and event review.

Objective: Detect events, collect reviewed examples and adapt classification to the camera context.

Engineering and implementation

- Reviewed examples form a training dataset; transfer learning and callbacks organize training and checkpoints.
- Temporal voting and a persistent alert outbox connect detection to operational notifications.

Technologies and methods: Python · OpenCV · Ultralytics/YOLO · PyTorch · Flask · RTSP

Mocap — browser inference and visualization

Independent projects, prototypes and research

Developed browser pose-estimation prototypes integrating inference, keypoint visualization and temporal smoothing to explore motion capture and 3D interaction.

Application area: Human movement, pose capture and avatar interaction.

Users and context: Body interaction and movement visualization experiments.

Objective: Estimate pose in video and connect body landmarks to browser visualization.

Application area: Computer vision and movement analysis.

Users and context: Video and body representation experimentation.

Objective: Extract and visualize skeletons from video.

Engineering and implementation

- Existing pose models process video and webcam input in the browser.
- Keypoint rendering, smoothing and 3D resources connect inference to interactive visualization.

Technologies and methods: JavaScript · TensorFlow.js · MoveNet · BlazePose · MediaPipe · WebGL

Technical cases — agricultural and business applications

PLANAGRO — inputs, applications and agronomic rules

Professional experience — Agrodan

Developed agricultural planning components in PLANAGRO, translating input application, dosage, withholding periods, drift and budgets into rules and

interfaces connected to operational data.

Application area: Agribusiness, field operations, packing, quality, logistics and management.

Users and context: Agrodan field, operational, administrative and management teams.

Objective: Integrate operational records, business rules and indicators to support execution and management.

Engineering and implementation

- Domain rules represent dosage, withholding periods, drift and application constraints.
- Planning interfaces connect operational inputs and budget information to the application workflow.

Technologies and methods: React · JavaScript · Node.js · operational APIs · agronomic rules

Irrigation — scheduling, genetic algorithms and Kc modeling

Professional experience — Agrodan

Explored irrigation scheduling, genetic algorithms and neural modeling of crop coefficient Kc, working with variable encoding, normalization, network parameters and operational backend integration.

Application area: Agribusiness, field operations, packing, quality, logistics and management.

Users and context: Agrodan field, operational, administrative and management teams.

Objective: Integrate operational records, business rules and indicators to support execution and management.

Engineering and implementation

- Categorical encoding and normalization prepare variables such as farm, phase, month and soil.
- Neural training, serialized model restoration and evolutionary experiments explore irrigation-related decisions.

Technologies and methods: JavaScript · Node.js · genetic algorithms · neural network · normalization

Spraying audit — maps and field execution

Professional experience — Agrodan

Built spraying-audit components using spatial and operational data, relating trajectories, speed, field blocks and services to support investigation of field execution.

Application area: Agribusiness, field operations, packing, quality, logistics and management.

Users and context: Agrodan field, operational, administrative and management teams.

Objective: Integrate operational records, business rules and indicators to support execution and management.

Engineering and implementation

- Maps connect movement and speed to field blocks and operational records.
- Audit interfaces make execution patterns available for operational review.

Technologies and methods: React · geospatial data · trajectory · speed · operational APIs

Dry matter — harvest forecasting and readiness

Professional experience — Agrodan

Implemented dry-matter analysis and harvest-forecast components connecting readings, varieties and parameters to charts and monitoring rules.

Application area: Agribusiness, field operations, packing, quality, logistics and management.

Users and context: Agrodan field, operational, administrative and management teams.

Objective: Integrate operational records, business rules and indicators to support execution and management.

Engineering and implementation

- Equations and parameters relate dry-matter observations to harvest readiness.
- Charts and operational interfaces support monitoring by variety and field context.

Technologies and methods: React · JavaScript · curves and equations · charts · field data

Field notebook and purchasing — local data and synchronization

Professional experience — Agrodan

Developed field-data collection and purchase-request workflows with local storage, process components and synchronization routines integrated with domain-specific backend services.

Application area: Agribusiness, field operations, packing, quality, logistics and management.

Users and context: Agrodan field, operational, administrative and management teams.

Objective: Integrate operational records, business rules and indicators to support execution and management.

Engineering and implementation

- Dexie/IndexedDB stores field and purchasing data locally.
- Synchronization handlers and domain APIs connect local records to the backend.

Technologies and methods: React · Dexie · IndexedDB · Node.js · synchronization

UGB management, people and improvement suggestions

Professional experience — Agrodan

Modeled business-unit and people-management applications and improvement-suggestion workflows combining organizational records, history, assessments, roles and documents.

Application area: Agribusiness, field operations, packing, quality, logistics and management.

Users and context: Agrodan field, operational, administrative and management teams.

Objective: Integrate operational records, business rules and indicators to support execution and management.

Engineering and implementation

- Business units, employees and roles share domain models and reusable components.
- Assessments, history, 5S and improvement workflows organize management information.

Technologies and methods: React · Node.js · organizational modeling · assessments · history

Commercial monitoring, forecasts and pallet movements

Professional experience — Agrodan

Developed commercial and logistics components connecting records and forecasts to management interfaces and operational pallet/container data.

Application area: Agribusiness, field operations, packing, quality, logistics and management.

Users and context: Agrodan field, operational, administrative and management teams.

Objective: Integrate operational records, business rules and indicators to support execution and management.

Engineering and implementation

- Commercial entries, forecasts and exchange-rate information support operational views.
- Pallet and container movements link logistics records to management interfaces.

Technologies and methods: React · Node.js · commercial data · pallets · containers

Technical cases — data engineering

Modular ETL and operational data quality

Professional experience — Agrodan

Structured integration pipelines with reusable importers, batch processing and domain rules for production, packing, quality and people, organizing operational data for analysis and decisions.

Application area: Agribusiness, field operations, packing, quality, logistics and management.

Users and context: Agrodan field, operational, administrative and management teams.

Objective: Integrate operational records, business rules and indicators to support execution and management.

Engineering and implementation

- Node.js and SQL importers separate extraction, transformation and domain-specific processing.
- Batch persistence, joins and aggregation connect ERP and operational records to indicators.

Technologies and methods: Node.js · SQL · Oracle · Sequelize · ETL · batch import

Technical cases — mobile and enterprise web

SIAgro — mobile operational data collection

Professional experience — Agrodan

Developed SIAgro mobile modules for work orders, production records and quality using React Native/Expo, business-object modeling, local persistence and operational indicators.

Application area: Agribusiness, field operations, packing, quality, logistics and management.

Users and context: Agrodan field, operational, administrative and management teams.

Objective: Integrate operational records, business rules and indicators to support execution and management.

Engineering and implementation

- Mobile workflows organize work orders, employees, targets and quality records.
- Reusable local persistence and API integration connect collection and dashboard components.

Technologies and methods: React Native · Expo · AsyncStorage · React Navigation · APIs

Hospital indicators and academic gamification

Hospital internship and university project

During the hospital internship, developed a Java EE web system for indicators and assessments. At university, built a project-management application with gamification and persistence, exploring progression rules and task organization.

Application area: Healthcare, hospital management and strategic planning.

Users and context: Hospital Universitário de Petrolina teams and managers.

Objective: Organize indicators, assessments and hospital planning monitoring.

Application area: Project management, gamification and collaboration.

Users and context: Project teams; university development project.

Objective: Represent projects, stages and progression through quests, experience and rewards.

Engineering and implementation

- EBSERH: Java EE, EJB, JSF, PrimeFaces and JPA/Hibernate for hospital indicators and assessments.
- The Game Project: persistent tasks, progression rules, scheduling and an interactive project timeline.

Technologies and methods: Java EE · EJB · JSF · PrimeFaces · JPA/Hibernate · JPQL · Quartz

Project links

Public repository — EBSERH · Public repository — The Game Project

Technical cases — decision support

Utilibox — multicriteria analysis, queues and applied methods

Independent projects, prototypes and research

Implemented decision-support and learning tools using multicriteria methods, queueing theory, simulation, financial mathematics and spaced repetition, translating models into explainable interfaces and calculations.

Application area: Finance, employment, production engineering, decision making, education, writing, health, beauty, documents, design and marketing.

Users and context: General users, students, writers and professionals.

Objective: Turn calculations, methods and knowledge organization into accessible interactive tools.

Engineering and implementation

- AHP, TOPSIS and PROMETHEE connect criteria and alternatives to decision analysis.
- Queueing, financial and spaced-repetition tools apply domain methods in interactive software.

Technologies and methods: Astro · JavaScript · TypeScript · AHP · TOPSIS · PROMETHEE · simulation

Project links

Open tool — ahp-online · Open tool — topsis · Open tool — promethee · Open tool — queueing-theory · Open tool — vsm · Open tool — anki-flashcards

Cartola and logistics — decisions under constraints

Independent projects, prototypes and research

Developed pipelines connecting data and optimization to decisions, exploring team selection under budget/formation constraints and routing with geographical visualization.

Application area: Sports, fantasy football, probabilistic forecasting and optimization.

Users and context: Performance research and Cartola lineup construction.

Objective: Estimate performance and select lineups under budget and other constraints.

Application area: Logistics, routing and geospatial analysis.

Users and context: Travel planning and logistics operations.

Objective: Represent locations and routes on maps and support routing decisions.

Engineering and implementation

- Cartola combines performance data and probabilistic estimates with constrained team selection.
- Logistics prototypes combine routes, maps and optimization experiments.

Technologies and methods: Python · pandas · Parquet · OR-Tools/CBC · React · Leaflet

Project links

Public repository — Cartola Oracle

Technical cases — products and knowledge

Narra and UtiliNotas — domain and knowledge applications

Independent projects, prototypes and research

Built narrative and knowledge-organization modules with entity relationships, contextual assistance, response traceability and note interoperability.

Application area: Finance, employment, production engineering, decision making, education, writing, health, beauty, documents, design and marketing.

Users and context: General users, students, writers and professionals.

Objective: Turn calculations, methods and knowledge organization into accessible interactive tools.

Engineering and implementation

- Related entities and linked notes organize narrative and knowledge contexts.
- Contextual assistance and interoperability connect editing, information and responses.

Technologies and methods: JavaScript · TypeScript · Astro · entities and relationships · Markdown

Project links

Open tool — narra · Open tool — utilinotas

Marketplace, automation and 3D interaction

Independent projects, prototypes and research

Structured products and prototypes with access rules, integrations and supervised workflows, including a marketplace, commercial collection, personal finance and a narrative 3D experience.

Application area: Games, interactive narrative and 3D graphics.

Users and context: Interactive narrative experience exploration.

Objective: Build an exploration environment with interaction, physics and narrative elements.

Application area: Entrepreneurship, marketplaces and product validation.

Users and context: Product creators and potential users.

Objective: Connect product proposals, discovery and validation in a web platform.

Application area: Careers, recruitment and supervised automation.

Users and context: Professional opportunity organization.

Objective: Collect opportunities and rank fit under user supervision.

Application area: Personal finance, Open Finance and transaction analysis.

Users and context: Personal financial organization.

Objective: Integrate and analyze transactions to support financial monitoring.

Application area: Data collection, offers and content distribution.

Users and context: Offer monitoring and publishing operations.

Objective: Structure offer information collection, processing and distribution.

Engineering and implementation

- Product roles and integrations organize campaigns and commercial workflows.
- Supervised automation and interactive prototypes explore different product domains.

Technologies and methods: Next.js · Supabase · PostgreSQL · Stripe · Playwright · SQLite

Technical cases — infrastructure and desktop integration

IT operations management — Agrodan

Professional experience — Agrodan

Managed IT operations, integrating support, servers, networks, continuity, ERP and IoT initiatives with delivery of technology services and solutions.

Application area: Agribusiness, field operations, packing, quality, logistics and management.

Users and context: Agrodan field, operational, administrative and management teams.

Objective: Integrate operational records, business rules and indicators to support execution and management.

Engineering and implementation

- Professional responsibilities cover support, Windows/Linux administration, Active Directory and corporate networks.
- Backup/recovery, virtualization, monitoring, security and IoT belong to the declared IT management scope.

Technologies and methods: Windows Server · Linux · Active Directory · Zabbix · GLPI · networks · IoT

MetaTrader5 desktop connector — TraderMind

Independent projects, prototypes and research

Built a Python connector to the MetaTrader5 desktop terminal, integrating market/account data with the documented research and execution workflow.

Application area: AI, Machine Learning and MLOps for quantitative research.

Users and context: Financial-market model and strategy research and experimentation.

Objective: Structure the data, training, temporal validation, inference and model supervision lifecycle; integrate signals and policies with MetaTrader5.

Engineering and implementation

- The connector uses the MetaTrader5 package and requires an authenticated local terminal.
- Candles and account data connect policy research with separate API, dashboard and execution components.

Technologies and methods: Python · MetaTrader5 · Flask · SQLite · local APIs

Technical cases — 3D and spatial visualization

Dente de Leão — interactive narrative world

Independent projects, prototypes and research

Built a narrative 3D prototype with character controls, camera, interaction, physics and HUD, organizing state and rendering with React Three Fiber, Rapier and Zustand.

Application area: Games, interactive narrative and 3D graphics.

Users and context: Interactive narrative experience exploration.

Objective: Build an exploration environment with interaction, physics and narrative elements.

Engineering and implementation

- World, player, camera and interactions are organized into game modules.
- React Three Fiber, Rapier and Zustand integrate rendering, physics and application state.

Technologies and methods: TypeScript · React Three Fiber · Three.js · Rapier · Zustand

Maps, models and spatial exploration

Independent projects, prototypes and research

Explored spatial visualization and 3D models with Leaflet and Three.js, building map, drawing, routing and scene-interaction interfaces.

Application area: Maps, routing and geographic interfaces.

Users and context: Spatial information visualization and exploration.

Objective: Represent geographic data and routes in a web interface.

Application area: Logistics, routing and geospatial analysis.

Users and context: Travel planning and logistics operations.

Objective: Represent locations and routes on maps and support routing decisions.

Application area: Computer graphics and model visualization.

Users and context: 3D scene and movement studies.

Objective: Explore 3D model presentation and movement.

Application area: Computer graphics and 3D web development.

Users and context: Scene, model and interaction studies.

Objective: Explore browser rendering of scenes and 3D assets.

Engineering and implementation

- Logistics and Leaflet projects combine map interfaces and routing components.
- Three.js experiments include scenes, interaction and model loading using third-party libraries.

Technologies and methods: JavaScript · React · Leaflet · Three.js · Canvas · FBX

Technical cases — Globalcom Agentic AI solution architecture

Modern Concept · functional and technical architecture delivered; implementation in progress.

Contribution: functional and technical architecture design and delivery. Implementation is in progress; production operation and operational metrics have not yet been declared.

Architecture-specified stack: TypeScript · React · Node.js · Python · FastAPI · Supabase · PostgreSQL · pgvector · RAG · MCP · Playwright · RLS · RBAC · Queues and workers

Agentic AI architecture and specialist orchestration

Objective: Coordinate specialized tasks with evidence and human oversight

- Designed 10 specialist roles: compliance, evidence, documents, research, writing, communication, deadlines, risk, systems audit and integration/ RPA.
- A dedicated orchestrator checks preconditions, distributes tasks and consolidates results; an independent auditor reviews agents, tools, sources and decisions.
- Execution contracts connect school, process, requirement and task context to approved tools, evidence, confidence and human escalation.

Managed RAG and document intelligence

Objective: Retrieve knowledge matching school, requirement and document version

- Designed six collections: official regulations, operational knowledge, management context, email precedents, school-specific documents and approved templates.
- Hybrid retrieval architecture with embeddings and pgvector; tenant, school, framework, requirement, version, validity and approval filters.
- Specified ingestion, classification, extraction, versioning, embedding reprocessing, chunk lineage and citation persistence.

Workflow, events and persistent execution

Objective: Keep processes and tasks running without an active user session

- The specified state machine controls dependencies, owners, deadlines, preconditions and completion criteria across nine process phases.
- Designed queues, events, webhooks and scheduled jobs, with specialist workflow, agent, document, RAG, notification, RPA and audit workers.
- Suggested Node.js/TypeScript, NestJS or Fastify, Zod and BullMQ or an alternative queue; Temporal is an optional durable workflow engine.

AI governance, security and continuous audit

Objective: Control autonomy, data access and decision quality

- The specified execution policy separates automatic execution, human approval actions and critical decisions reserved for people.
- Designed partner/school isolation with RLS and RBAC, least privilege, secret management, logs, retention, backup and recovery.
- An independent auditor checks citation/evidence alignment, regulation versions, data integrity and workflow compliance; quarterly audits and four-month review meetings are planned.

API, MCP and RPA integrations for external operations

Objective: Connect sources and portals with controlled execution

- Integration strategy prioritizes APIs, uses MCP for standardized tool access and proposes RPA when formal integration is unavailable.

- Specified isolated Playwright browser workers, retry policies, idempotency, layout change detection and manual fallback.
- Architectural scope includes Notion migration/synchronization, email, portals and future school/financial system connectors; QuickSchools, QuickBooks and Wave are initial references.

Certification domain and product modeling

Objective: Translate certification operations into product entities and workflows

- Delivered domain design connects partners, schools, frameworks, phases, requirements, tasks, documents, evidence, risks, decisions and history.
- Supabase/PostgreSQL is the designed canonical store, with document Storage, Auth, Realtime and pgvector; a React/TypeScript frontend includes workflow visualization.
- Architecture covers readiness, gaps, corrective plans, self-study and artifact reuse; expected outcomes are product goals rather than achieved impact metrics.

Consulting, solution architecture and roadmap

Objective: Deliver the architecture and roadmap for incremental implementation

- Documented deliverables combine an executive view, detailed architecture, navigable PT/EN presentation and implementation planning.
- Roadmap starts with discovery and operational domain, then adds RAG/documents, agents, continuous compliance, integrations and certification intelligence.
- Defined target measures: phase duration, evidence completeness, reuse, draft approval, agent errors, integration health and audit coverage.

Technical cases — TraderMind MLOps and model lifecycle

Data pipelines, features and quality contracts

TraderMind · independent MLOps and quantitative research project

Integrated extraction, data updates, dataset construction and feature selection into operational TraderMind pipeline stages.

Application area: AI, Machine Learning and MLOps for quantitative research.

Users and context: Financial-market model and strategy research and experimentation.

Objective: Structure the data, training, temporal validation, inference and model supervision lifecycle; integrate signals and policies with MetaTrader5.

Engineering and implementation

- Market data and multitimeframe indicators, Parquet datasets and evolutionary feature selection with EFS-AHP.
- Stage checks cover temporal integrity, feature availability and health, configuration consistency and artifact synchronization.

Technologies and methods: Python · pandas · NumPy · Parquet · DEAP · EFS-AHP

Training, checkpoints and artifact management

TraderMind · independent MLOps and quantitative research project

Organized experiments, training and execution recovery while preserving the relationship between models, features and parameters.

Application area: AI, Machine Learning and MLOps for quantitative research.

Users and context: Financial-market model and strategy research and experimentation.

Objective: Structure the data, training, temporal validation, inference and model supervision lifecycle; integrate signals and policies with MetaTrader5.

Engineering and implementation

- Classifier training and comparison routines, feature selection and entry and early-exit models.
- Checkpoints resume evolutionary search; serialized models, feature lists, configuration and logs support workflow reproduction.

Technologies and methods: Python · scikit-learn · XGBoost · LightGBM · CatBoost · joblib · checkpoints

Temporal validation, leakage and out-of-sample evaluation

TraderMind · independent MLOps and quantitative research project

Separated selection, calibration and temporal evaluation to compare models and policies without contaminating the final evaluation period.

Application area: AI, Machine Learning and MLOps for quantitative research.

Users and context: Financial-market model and strategy research and experimentation.

Objective: Structure the data, training, temporal validation, inference and model supervision lifecycle; integrate signals and policies with MetaTrader5.

Engineering and implementation

- Walk-forward, temporal holdouts, backtests and leakage audits complement training; causal external-data alignment avoids future information.
- Policies are selected before final OOS testing; comparisons between logged operation and replay/backtest help investigate discrepancies.

Technologies and methods: Python · temporal validation · walk-forward · OOS · backtesting

Model serving and MetaTrader5 integration

TraderMind · independent MLOps and quantitative research project

Connected ML artifacts, an inference service and the desktop terminal to bridge research and demonstration operation.

Application area: AI, Machine Learning and MLOps for quantitative research.

Users and context: Financial-market model and strategy research and experimentation.

Objective: Structure the data, training, temporal validation, inference and model supervision lifecycle; integrate signals and policies with MetaTrader5.

Engineering and implementation

- A Flask API loads models and their features by pair/direction, returns predictions and probabilities and supports exit checks.
- A Python connector integrates data and execution with MetaTrader5; service endpoints expose health, status, loaded models and uptime.

Technologies and methods: Python · Flask · REST APIs · joblib · MetaTrader5

Orchestration, retraining and operational updates

TraderMind · independent MLOps and quantitative research project

Coordinated data, training, evaluation and serving cycles through distinct execution modes.

Application area: AI, Machine Learning and MLOps for quantitative research.

Users and context: Financial-market model and strategy research and experimentation.

Objective: Structure the data, training, temporal validation, inference and model supervision lifecycle; integrate signals and policies with MetaTrader5.

Engineering and implementation

- The supervisor organizes extraction, datasets, EFS-AHP, training, backtesting and serving; it distinguishes data updates without retraining from full retraining cycles.
- Runs use identifiers and logs; scheduling and operational checks support pipeline maintenance.

Technologies and methods: Python · supervisor · scheduling · logs · pipeline automation

Observability and model follow-up

TraderMind · independent MLOps and quantitative research project

Connected the operational dashboard, pipeline stage health and performance review using recent data.

Application area: AI, Machine Learning and MLOps for quantitative research.

Users and context: Financial-market model and strategy research and experimentation.

Objective: Structure the data, training, temporal validation, inference and model supervision lifecycle; integrate signals and policies with MetaTrader5.

Engineering and implementation

- Dashboard and health records track stages, training and operation; APIs expose model and service availability.
- OOS window updates and comparisons between logged trades and backtests support regime-change analysis and model review decisions.

Technologies and methods: Python · Flask · dashboard · logs · OOS · monitoring

Technical cases — Agrodan industrial computer vision platform

Computer vision platform development for packing operations

Agrodan · Agritech.NE applied R&D project · platform developer

Worked as a developer on the platform project during my Agrodan employment period, connecting fruit counting and tracking to operational packing monitoring.

Application area: Agribusiness, fruit production, packing, computer vision, traceability and operational efficiency.

Users and context: Sorting-machine operators, packing teams and Agrodan managers.

Objective: Integrate fruit counts and industrial data to monitor batches, indicate transitions and reduce interruptions.

Engineering and implementation

- Platform for identifying and recording mangoes on the reject conveyor, with counting and tracking to avoid duplicates.
- Connected the counting application, industrial data API, dashboard and notifications; the report describes two operating cameras.
- Applied R&D project with Agrodan in the Agritech.NE context, using CRISP-DM and project result follow-up.

Technologies and methods: Computer vision · Deep Learning · Cameras · APIs · ERP · Sorting-machine integration · Dashboard · Telegram · Dedicated server · CRISP-DM

ERP, sorting-machine integration and batch traceability

Agrodan · Agritech.NE applied R&D project · platform developer

The platform connects incoming batches, packed volume and rejected fruit to track fruit origin and support batch transitions.

Application area: Agribusiness, fruit production, packing, computer vision, traceability and operational efficiency.

Users and context: Sorting-machine operators, packing teams and Agrodan managers.

Objective: Integrate fruit counts and industrial data to monitor batches, indicate transitions and reduce interruptions.

Engineering and implementation

- API access to sorting-machine and packing data, combined with ERP batch records and variety, harvest and handling information.
- Fruit counts are combined with the sorting-machine size distribution and variety-specific weight ranges to estimate reject weight.
- Reconciled incoming volume, packed volume and estimated reject weight; a 98% incoming-volume proximity threshold supports batch-change decisions.

Technologies and methods: Computer vision · Deep Learning · Cameras · APIs · ERP · Sorting-machine integration · Dashboard · Telegram · Dedicated server · CRISP-DM

Dashboard, alerts and operational decision support

Agrodan · Agritech.NE applied R&D project · platform developer

Developed the monitoring and notification platform to provide operators with processing information and batch-change timing.

Application area: Agribusiness, fruit production, packing, computer vision, traceability and operational efficiency.

Users and context: Sorting-machine operators, packing teams and Agrodan managers.

Objective: Integrate fruit counts and industrial data to monitor batches, indicate transitions and reduce interruptions.

Engineering and implementation

- Monitored integrated production data and prepared Telegram alerts when processed volume approaches incoming volume.
- The scope includes dedicated-server counting and monitoring applications; the report records training and follow-up in February 2024.
- Counts, size distributions and processing volume support operational decisions beyond manual fruit removal from the tank.

Technologies and methods: Computer vision · Deep Learning · Cameras · APIs · ERP · Sorting-machine integration · Dashboard · Telegram · Dedicated server · CRISP-DM

Project outcomes and industrial process improvement

Agrodan · Agritech.NE applied R&D project · platform developer

Platform integration into the packing workflow produced project-level outcomes recorded in the final report, covering throughput, downtime and operating costs.

Application area: Agribusiness, fruit production, packing, computer vision, traceability and operational efficiency.

Users and context: Sorting-machine operators, packing teams and Agrodan managers.

Objective: Integrate fruit counts and industrial data to monitor batches, indicate transitions and reduce interruptions.

Engineering and implementation

- Reported throughput increased from 30 to 37.5 boxes per hour; daily control-change downtime decreased from 45 to 5 minutes.
- The report records a 17% reduction in operating costs associated with labor and processing time.
- Reported counting precision was 98%; variety-specific calibration remained necessary, with a Kent mango adjustment identified in the report.

Technologies and methods: Computer vision · Deep Learning · Cameras · APIs · ERP · Sorting-machine integration · Dashboard · Telegram · Dedicated server · CRISP-DM

Doctoral, hospital internship and university projects

EBSERH — hospital indicators and assessments

Java EE web system for hospital indicators, assessments and strategic planning, associated with the Hospital Universitário de Petrolina internship.

Java, Java EE, EJB, JPA, Hibernate, JSF, PrimeFaces, XHTML, JPQL, Apache Ant, NetBeans

Public repository — EBSERH

The Game Project — academic project management

University application connecting persistent tasks, gamification and an interactive project timeline.

Java, JSF, PrimeFaces, JPA, Hibernate, MySQL, JPQL, Quartz, XHTML, JavaScript, jQuery, Apache Ant, NetBeans

Public repository — The Game Project

EFS-AHP — feature selection and multicriteria research

Doctoral research combining feature selection, evolutionary algorithms, AHP and comparative ML experiments to study predictive quality, dimensionality and training cost.

JavaScript, Node.js, Python, Jupyter, Random Forest, LASSO, PCA, t-SNE, UMAP, AHP, Docker, pandas, NumPy, scikit-learn, SciPy, DEAP

Public repository — EFS-AHP

SIA — business modules

94 business modules and submodules across 23 local Git references.

These entries belong to a shared ecosystem and may overlap across interfaces, APIs, data processing and integrations. They are not a count of independent products.

Fleet refueling

Application: Integration of information for fleet refueling.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Absenteeism

Application: Integration of information for absenteeism.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Operational follow-up

Application: Operational workflow for operational follow-up.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @material-ui/lab, @nivo/bar, @nivo/pie, color-interpolate, dexie, react, react-csv, react-custom-scrollbars, react-redux, thenby

- IndexedDB persistence through Dexie
- Local data layer
- Synchronization references
- State hooks
- Effect hooks and data loading

Field and packing areas

Application: Operational workflow for field and packing areas.

Professional context — Agrodan

Observed stack: @material-ui/core, @nivo/bar, moment, react, react-router-dom

- State hooks

Atel Escola monitoring

Application: Operational workflow for atel escola monitoring.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, react-router-dom, turf

- Maps and spatial information
- Leaflet maps

Spraying audit

Application: Operational workflow for spraying audit.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, @use-it/event-listener, crypto-js, leaflet, moment, react, react-bootstrap, react-dom, react-leaflet, react-leaflet-canvas-layer, react-leaflet-multi-options-polyline, react-router-dom, turf

- Maps and spatial information
- Leaflet maps
- Corresponding backend routes located

SGA audit

Application: Integration of information for sga audit.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

SIA audit

Application: Operational workflow for sia audit.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, moment-duration-format, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Team assessment

Application: Operational workflow for team assessment.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, dexie, react, react-select

- IndexedDB persistence through Dexie
- Local data layer
- Synchronization handlers
- Synchronization references
- State hooks
- Corresponding backend routes located

Administrative team assessment

Application: Operational workflow for administrative team assessment.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, dexie, react, react-select

- IndexedDB persistence through Dexie
- Local data layer
- Synchronization handlers
- Synchronization references
- State hooks

Student–teacher assessment

Application: Operational workflow for student–teacher assessment.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, dexie, react, react-select

- IndexedDB persistence through Dexie
- Local data layer
- Synchronization handlers
- Synchronization references
- State hooks

Manager–manager assessment

Application: Operational workflow for manager–manager assessment.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, dexie, react, react-select

- IndexedDB persistence through Dexie
- Local data layer
- Synchronization handlers
- Synchronization references
- State hooks

Teacher–manager assessment

Application: Operational workflow for teacher–manager assessment.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, dexie, react, react-select

- IndexedDB persistence through Dexie
- Local data layer
- Synchronization handlers
- Synchronization references
- State hooks

Teacher–teacher assessment

Application: Operational workflow for teacher–teacher assessment.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, dexie, react, react-select

- IndexedDB persistence through Dexie
- Local data layer
- Synchronization handlers
- Synchronization references
- State hooks

Reverse assessment

Application: Operational workflow for reverse assessment.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, dexie, react, react-select

- IndexedDB persistence through Dexie
- Local data layer
- Synchronization handlers
- Synchronization references
- State hooks
- Corresponding backend routes located

Peer assessment

Application: Operational workflow for peer assessment.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, dexie, react, react-select

- IndexedDB persistence through Dexie
- Local data layer
- Synchronization handlers
- Synchronization references
- State hooks
- Corresponding backend routes located

Time-clock records

Application: Integration of information for time-clock records.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

CFO view

Application: Integration of information for cfo view.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Commercial results

Application: Operational workflow for commercial results.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, dexie, fs, moment, pdfreader, react, react-file-base64, react-select

- IndexedDB persistence through Dexie
- Local data layer
- Synchronization handlers
- Synchronization references
- State hooks
- Effect hooks and data loading
- Corresponding backend routes located

Accounts payable

Application: Integration of information for accounts payable.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Field-block registration

Application: Integration of information for field-block registration.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Field notebook

Application: Operational workflow for field notebook.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @mui/material, dexie, moment, react, react-chartjs-2, react-select, uuidv4

- IndexedDB persistence through Dexie
- Local data layer
- Synchronization handlers
- Synchronization references
- State hooks
- Effect hooks and data loading
- Corresponding backend routes located

Packed boxes

Application: Integration of information for packed boxes.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Packed boxes by business unit

Application: Integration of information for packed boxes by business unit.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Input verification

Application: Integration of information for input verification.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Pallet control

Application: Integration of information for pallet control.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Quality control

Application: Integration of information for quality control.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Receiving control

Application: Operational workflow for receiving control.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- State hooks
- Effect hooks and data loading
- Maps and spatial information
- Leaflet maps

Harvest schedule

Application: Operational workflow for harvest schedule.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, react-touch-screen-keyboard, turf

- Maps and spatial information
- Leaflet maps

Harvest schedule form

Application: Operational workflow for harvest schedule form.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, dexie, moment, react, react-touch-screen-keyboard

- IndexedDB persistence through Dexie
- Local data layer
- Synchronization references

SIA-integrated harvest schedule

Application: Operational workflow for sia-integrated harvest schedule.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, dexie, moment, react

- IndexedDB persistence through Dexie
- Local data layer
- Synchronization references
- State hooks

- Effect hooks and data loading
- Corresponding backend routes located

Field sketches

Application: Operational workflow for field sketches.

Professional context — Agrodan

Observed stack: @material-ui/core, @nivo/pie, moment, react

Costs

Application: Integration of information for costs.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Satisfaction dashboard

Application: Integration of information for satisfaction dashboard.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Packaging

Application: Operational workflow for packaging.

Professional context — Agrodan

Observed stack: @material-ui/core, @nivo/bar, moment, react

Virtual inventory

Application: Integration of information for virtual inventory.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Billing

Application: Integration of information for billing.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Field-block closure

Application: Integration of information for field-block closure.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Price fluctuations

Application: Integration of information for price fluctuations.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Payroll

Application: Integration of information for payroll.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Inventory management

Application: Integration of information for inventory management.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Business-unit management

Application: Operational workflow for business-unit management.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @material-ui/styles, @react-pdf/renderer, axios, dexie, file-saver, moment, prop-types, react, react-icons, react-pdf, react-router-dom, react-select, react-to-pdf, xlsx

- Synchronization references
- State hooks
- Effect hooks and data loading
- Spreadsheets
- HTTP integration
- Corresponding backend routes located

Hour meter

Application: Integration of information for hour meter.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

BPM indicators and satisfaction

Application: Operational workflow for bpm indicators and satisfaction.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, dexie, leaflet, moment, prop-types, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf, uuidv4

- IndexedDB persistence through Dexie
- Local data layer
- Synchronization references

- Maps and spatial information
- Leaflet maps
- Corresponding backend routes located

Logistics satisfaction

Application: Operational workflow for logistics satisfaction.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, moment, react

IT indicators and ticket queue

Application: Operational workflow for it indicators and ticket queue.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps
- Corresponding backend routes located

Irrigation monitoring

Application: Operational workflow for irrigation monitoring.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, moment-duration-format, react, react-csv-reader, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps
- Corresponding backend routes located

Irrigas view

Application: Integration of information for irrigas view.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Lean Packing Game

Application: Integration of information for lean packing game.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Time recording

Application: Operational workflow for time recording.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, moment, react, react-chartjs-2, trabalho-eficiente.jpg

- Corresponding backend routes located

MAD monitoring

Application: Operational workflow for mad monitoring.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, react-router-dom, turf

- Local data layer
- Maps and spatial information
- Leaflet maps
- Corresponding backend routes located

MAF and mango monitoring

Application: Operational workflow for maf and mango monitoring.

Professional context — Agrodan

Observed stack: @material-ui/core, @nivo/bar, moment, react, socket.io-client

- State hooks
- Effect hooks and data loading
- Socket events
- Corresponding backend routes located

Integrated pest monitoring

Application: Operational workflow for integrated pest monitoring.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Local data layer
- Maps and spatial information
- Leaflet maps

Dry matter and harvest forecasting

Application: Operational workflow for dry matter and harvest forecasting.

Professional context — Agrodan

Observed stack: @material-ui/core, @nivo/pie, moment, react

- Forecasting function
- Harvest-readiness rule
- Corresponding backend routes located

Materials

Application: Operational workflow for materials.

Professional context — Agrodan

Observed stack: @material-ui/core, @nivo/pie, moment, react

- Corresponding backend routes located

Material movements

Application: Operational workflow for material movements.

Professional context — Agrodan

Observed stack: @material-ui/core, @nivo/pie, moment, react

- Corresponding backend routes located

Pallet and container movements

Application: Operational workflow for pallet and container movements.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, dexie, moment, react, react-touch-screen-keyboard

- IndexedDB persistence through Dexie
- Local data layer
- Synchronization references
- State hooks
- Corresponding backend routes located

Budget

Application: Integration of information for budget.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps
- Corresponding backend routes located

Organization chart and areas

Application: Operational workflow for organization chart and areas.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, dexie, react

- IndexedDB persistence through Dexie
- Synchronization references
- State hooks

Labor PDCA

Application: Operational workflow for labor pdca.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, dexie, moment, react, react-file-base64, react-select

- IndexedDB persistence through Dexie
- Local data layer
- Synchronization handlers
- Synchronization references
- State hooks
- Effect hooks and data loading

Packing and box handling

Application: Operational workflow for packing and box handling.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/bar, audio-loader, audio-play, moment, play-sound, react, react-router-dom, socket.io-client

- Socket events

PLANAGRO — agricultural planning and rules

Application: Operational workflow for planagro — agricultural planning and rules.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, dexie, leaflet, moment, path, prop-types, react, react-custom-scrollbars, react-data-table-component, react-leaflet, react-leaflet-canvas-layer, react-leaflet-control, react-select, react-to-pdf, react-to-print, recharts, turf, uuidv4, xlsx

- IndexedDB persistence through Dexie
- Local data layer

- Synchronization references
- State hooks
- Effect hooks and data loading
- Maps and spatial information
- Leaflet maps
- Spreadsheets
- Printing
- Withholding-period rule
- Dosage rule
- Drift rule
- Corresponding backend routes located

Results and exchange-rate forecasts

Application: Operational workflow for results and exchange-rate forecasts.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, dexie, moment, prop-types, react, react-select, uuidv4, xlsx

- IndexedDB persistence through Dexie
- Local data layer
- Synchronization references
- State hooks
- Spreadsheets
- Corresponding backend routes located

Processing schedule

Application: Integration of information for processing schedule.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Projects

Application: Integration of information for projects.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Employee counts

Application: Integration of information for employee counts.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Field quality and checklists

Application: Operational workflow for field quality and checklists.

Professional context — Agrodan

Observed stack: @material-ui/core, dexie, react, react-select

- IndexedDB persistence through Dexie
- Synchronization references
- State hooks
- Corresponding backend routes located

HR view

Application: Integration of information for hr view.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Occupational safety risks

Application: Operational workflow for occupational safety risks.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Phone extensions

Application: Integration of information for phone extensions.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Rejects

Application: Integration of information for rejects.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

BI reports

Application: Integration of information for bi reports.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, react-router-dom, turf

- Maps and spatial information
- Leaflet maps

Labor summary

Application: Integration of information for labor summary.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Management routines

Application: Integration of information for management routines.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Quality management system

Application: Operational workflow for quality management system.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, dexie, moment, react, react-file-base64, react-select

- IndexedDB persistence through Dexie
- Local data layer
- Synchronization handlers
- Synchronization references
- State hooks
- Effect hooks and data loading

Quality reports

Application: Operational workflow for quality reports.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, dexie, react, react-file-base64, react-select

- IndexedDB persistence through Dexie
- Local data layer
- Synchronization references
- State hooks
- Effect hooks and data loading

Time-bank balance

Application: Integration of information for time-bank balance.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Satisfaction

Application: Operational workflow for satisfaction.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, moment, react

Security

Application: Operational workflow for security.

Professional context — Agrodan

Observed stack: @material-ui/core, @nivo/bar, moment, react

Purchase requests

Application: Operational workflow for purchase requests.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, dexie, moment, react, react-select

- IndexedDB persistence through Dexie
- Local data layer
- Synchronization handlers
- Synchronization references
- State hooks
- Corresponding backend routes located

Improvement suggestions

Application: Operational workflow for improvement suggestions.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, axios, moment, react

- Synchronization references
- State hooks
- Effect hooks and data loading
- HTTP integration
- Corresponding backend routes located

Workshop business unit

Application: Integration of information for workshop business unit.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Harvest validation

Application: Integration of information for harvest validation.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

MDO validation

Application: Operational workflow for mdo validation.

Professional context — Agrodan

Observed stack: @material-ui/core, @nivo/pie, moment, react

- Corresponding backend routes located

Labor validation

Application: Integration of information for labor validation.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, turf

- Maps and spatial information
- Leaflet maps

Guard monitoring

Application: Operational workflow for guard monitoring.

Professional context — Agrodan

Observed stack: @material-ui/core, @material-ui/icons, @nivo/pie, leaflet, moment, react, react-dom, react-leaflet, react-leaflet-canvas-layer, react-router-dom, turf

- Maps and spatial information
- Leaflet maps

UGB — employees and history

Application: Operational workflow for ugb — employees and history.

Professional context — Agrodan

Observed stack: React, JavaScript

- Business-unit management workflow

UGB — units and roles

Application: Operational workflow for ugb — units and roles.

Professional context — Agrodan

Observed stack: React, JavaScript

- Business-unit management workflow

UGB — 5S

Application: Operational workflow for ugb — 5s.

Professional context — Agrodan

Observed stack: React, JavaScript

- Business-unit management workflow

UGB — people assessment

Application: Operational workflow for ugb — people assessment.

Professional context — Agrodan

Observed stack: React, JavaScript

- Business-unit management workflow

UGB — action plans

Application: Operational workflow for ugb — action plans.

Professional context — Agrodan

Observed stack: React, JavaScript

- Business-unit management workflow

UGB — technical qualification

Application: Operational workflow for ugb — technical qualification.

Professional context — Agrodan

Observed stack: React, JavaScript

- Business-unit management workflow

UGB — technical requalification

Application: Operational workflow for ugb — technical requalification.

Professional context — Agrodan

Observed stack: React, JavaScript

- Business-unit management workflow

UGB — improvement suggestions

Application: Operational workflow for ugb — improvement suggestions.

Professional context — Agrodan

Observed stack: React, JavaScript

- Business-unit management workflow

SIA / SIAgro — data and mobile workflows

48 technical workflows and modules across the checkout, ETL and mobile layers.

These entries belong to a shared ecosystem and may overlap across interfaces, APIs, data processing and integrations. They are not a count of independent products.

Production records

Application: Operational workflow for production records.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- Query table relationships

Operational shifts

Application: Operational workflow for operational shifts.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- Query table relationships

Business-unit inspections

Application: Operational workflow for business-unit inspections.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence

Pallet origin

Application: Operational workflow for pallet origin.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- SQL query aggregation

Production targets

Application: Operational workflow for production targets.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- Query table relationships

Target percentages

Application: Operational workflow for target percentages.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- SQL query aggregation

Yield by record and day

Application: Operational workflow for yield by record and day.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence

Yield updates

Application: Operational workflow for yield updates.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- SQL query aggregation

Total production

Application: Operational workflow for total production.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- SQL query aggregation

Production by farm

Application: Operational workflow for production by farm.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- SQL query aggregation
- Query table relationships

Business units and farms

Application: Operational workflow for business units and farms.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- Query table relationships

Packaging by business unit

Application: Operational workflow for packaging by business unit.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- Persisted collection reads
- SQL query aggregation
- Query table relationships

Daily packaging

Application: Operational workflow for daily packaging.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- Persisted collection reads
- SQL query aggregation
- Query table relationships

Field LUGB dataset

Application: Operational workflow for field lugb dataset.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- SQL query aggregation

Employee–business unit links

Application: Operational workflow for employee–business unit links.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- Persisted collection reads
- Query table relationships

Field production targets

Application: Operational workflow for field production targets.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- Persisted collection reads
- Query table relationships

Quality targets by business unit

Application: Operational workflow for quality targets by business unit.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- Persisted collection reads
- SQL query aggregation
- Query table relationships

Field target composition

Application: Operational workflow for field target composition.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- Persisted collection reads

- SQL query aggregation
- Query table relationships

Field NUGB

Application: Operational workflow for field nugb.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- Persisted collection reads

Quality by employee and day

Application: Operational workflow for quality by employee and day.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- Persisted collection reads
- SQL query aggregation
- Query table relationships

Field quality

Application: Operational workflow for field quality.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- Persisted collection reads
- SQL query aggregation
- Query table relationships

Quality breakdown

Application: Operational workflow for quality breakdown.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- Persisted collection reads

Packing quality

Application: Operational workflow for packing quality.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- Persisted collection reads
- Matching against specific records

Packing NUGB quality

Application: Operational workflow for packing nugb quality.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- Persisted collection reads
- Matching against specific records

Packing LUGB consolidation

Application: Operational workflow for packing lugb consolidation.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- Persisted collection reads
- Matching against specific records

Packing targets

Application: Operational workflow for packing targets.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- Persisted collection reads

APPT and Farmlog integration

Application: Operational workflow for appt and farmlog integration.

Professional context — Agrodan

Observed stack: Node.js, JavaScript, SQL, Sequelize

- Batch record persistence
- Persisted collection reads
- Query table relationships

Movement and access direction

Application: Operational workflow for movement and access direction.

Professional context — Agrodan

Observed stack: Node.js, Moment.js, SQL Server, Oracle

- Record grouping
- Batch record persistence

Hourly access distribution

Application: Operational workflow for hourly access distribution.

Professional context — Agrodan

Observed stack: Node.js, Moment.js, SQL Server, Oracle

- Record grouping
- Batch record persistence

Operational availability

Application: Operational workflow for operational availability.

Professional context — Agrodan

Observed stack: Node.js, Moment.js, SQL Server, Oracle

- Record grouping
- Batch record persistence

Turnstile utilization

Application: Operational workflow for turnstile utilization.

Professional context — Agrodan

Observed stack: Node.js, Moment.js, SQL Server, Oracle

- Record grouping
- Batch record persistence

Visitors and movement by department

Application: Operational workflow for visitors and movement by department.

Professional context — Agrodan

Observed stack: Node.js, Moment.js, SQL Server, Oracle

- Record grouping
- Batch record persistence

Attendance frequency

Application: Operational workflow for attendance frequency.

Professional context — Agrodan

Observed stack: Node.js, Moment.js, SQL Server, Oracle

- Record grouping
- Batch record persistence

Dynamic form builder

Application: Operational workflow for dynamic form builder.

Professional context — Agrodan

Observed stack: React, Redux, Material UI

Forms and themes API

Application: Operational workflow for forms and themes api.

Professional context — Agrodan

Observed stack: Express, Sequelize, SQL

Departments and subdepartments

Application: Operational workflow for departments and subdepartments.

Professional context — Agrodan

Observed stack: Node.js, Sequelize

Form completion orders

Application: Operational workflow for form completion orders.

Professional context — Agrodan

Observed stack: Sequelize, SQL

Weather ingestion

Application: Operational workflow for weather ingestion.

Professional context — Agrodan

Observed stack: Node.js, Sequelize

Operational routine scheduler

Application: Operational workflow for operational routine scheduler.

Professional context — Agrodan

Observed stack: Agenda, MongoDB, Agendash

- Scheduled job registration
- Scheduling interval definition

Harvest date re-estimation and actuals

Application: Operational workflow for harvest date re-estimation and actuals.

Professional context — Agrodan

Observed stack: Node.js, SQL, Sequelize

Validation and error traceability

Application: Operational workflow for validation and error traceability.

Professional context — Agrodan

Observed stack: JavaScript, Sequelize

- Batch record persistence

Mobile work orders

Application: Operational workflow for mobile work orders.

Professional context — Agrodan

Observed stack: React Native, Expo

- Query table relationships

Mobile production records

Application: Operational workflow for mobile production records.

Professional context — Agrodan

Observed stack: React Native, Expo

- In-app local storage

Employees in production records

Application: Operational workflow for employees in production records.

Professional context — Agrodan

Observed stack: React Native

- In-app local storage

Mobile quality

Application: Operational workflow for mobile quality.

Professional context — Agrodan

Observed stack: React Native, Expo

- In-app local storage

Targets and business units in work orders

Application: Operational workflow for targets and business units in work orders.

Professional context — Agrodan

Observed stack: React Native

- In-app local storage

Mobile operational dashboard

Application: Operational workflow for mobile operational dashboard.

Professional context — Agrodan

Observed stack: React Native, Victory Native

- In-app local storage

Reusable local persistence

Application: Operational workflow for reusable local persistence.

Professional context — Agrodan

Observed stack: React Native, AsyncStorage

- In-app local storage

Complete project and reference catalog

38 projects and references include professional applications, doctoral research, independent products, prototypes, studies, backups and third-party references. Source availability alone does not prove exclusive authorship, deployment or measured impact.

EBSERH — hospital indicators and assessments

Java EE web system for hospital indicators, assessments and strategic planning, associated with the Hospital Universitário de Petrolina internship.

Application area: Healthcare, hospital management and strategic planning.

Users and context: Hospital Universitário de Petrolina teams and managers.

Objective: Organize indicators, assessments and hospital planning monitoring.

Architecture: Java EE application with entities, persistence, business services and a JSF interface for indicators and assessments.

Stack: Java, Java EE, EJB, JPA, Hibernate, JSF, PrimeFaces, XHTML, JPQL, Apache Ant, NetBeans

Public repository — EBSERH

The Game Project — academic project management

University application connecting persistent tasks, gamification and an interactive project timeline.

Application area: Project management, gamification and collaboration.

Users and context: Project teams; university development project.

Objective: Represent projects, stages and progression through quests, experience and rewards.

Architecture: Web application with entities, data access, interface controllers and MySQL persistence; gamification connects adventures, battles and progression.

Stack: Java, JSF, PrimeFaces, JPA, Hibernate, MySQL, JPQL, Quartz, XHTML, JavaScript, jQuery, Apache Ant, NetBeans

Public repository — The Game Project

EFS-AHP — feature selection and multicriteria research

Doctoral research combining feature selection, evolutionary algorithms, AHP and comparative ML experiments to study predictive quality, dimensionality and training cost.

Application area: Data science, Machine Learning and multicriteria decision making.

Users and context: Applied doctoral research in Software Engineering.

Objective: Select features and compare model alternatives by performance, dimensionality and cost.

Architecture: Experiments organized by datasets and methods, with evolutionary routines, multicriteria evaluation, statistical notebooks and a containerized environment.

Stack: JavaScript, Node.js, Python, Jupyter, Random Forest, LASSO, PCA, t-SNE, UMAP, AHP, Docker, pandas, NumPy, scikit-learn, SciPy, DEAP

Public repository — EFS-AHP

Agrodan — operational application components and data ecosystem

Professional SIA/SIAgro ecosystem connecting agricultural data, access and availability indicators, corporate forms, weather collection, Node.js/SQL services and React/React Native applications.

Application area: Agribusiness, field operations, packing, quality, logistics and management.

Users and context: Agrodan field, operational, administrative and management teams.

Objective: Integrate operational records, business rules and indicators to support execution and management.

Architecture: React/Redux frontend, modular Express backend, data integration and processing layers and a React Native/Expo application for SIAgro processes.

Stack: JavaScript, Node.js, Express, SQL, Sequelize, Oracle, oracledb, Moment.js, mssql, React, Redux, Nivo, Umzug, React Redux, Material UI, JSS, Jest / react-scripts, Axios, axios-https-proxy-fix, Agenda, MongoDB, Agendash, Cron, React Native, Expo, React Navigation, AsyncStorage, Victory Native, Galio Framework, Expo File System, Expo Image Manipulator

Genetic algorithms — irrigation and operations research

JavaScript laboratory separating functions, datasets and results to explore evolutionary optimization, irrigation and operations research.

Application area: Operations research and irrigation scheduling.

Users and context: Agricultural optimization studies.

Objective: Explore genetic algorithms for scheduling and constraint problems.

Architecture: Experimental solution representation, evolution and evaluation routines, with operations research formulations and dataset and result recording.

Stack: JavaScript, Node.js, GLPK.js, ml-confusion-matrix, Blessed

EFS-AHP research backup

Historical research archive containing experiments, models and statistical analyses; a version archive rather than an independent product.

Application area: Data science, Machine Learning and multicriteria decision making.

Users and context: Applied doctoral research in Software Engineering.

Objective: Select features and compare model alternatives by performance, dimensionality and cost.

Architecture: Experiments organized by datasets and methods, with evolutionary routines, multicriteria evaluation, statistical notebooks and a containerized environment.

Stack: JavaScript, Python, Jupyter, Docker, pandas, NumPy, scikit-learn, SciPy, DEAP

Cartola Oracle — probabilistic forecasts and constrained teams

Pipeline using performance histories, Markov representations and integer programming to select athletes under budget and formation constraints.

Application area: Sports, fantasy football, probabilistic forecasting and optimization.

Users and context: Performance research and Cartola lineup construction.

Objective: Estimate performance and select lineups under budget and other constraints.

Architecture: Pipeline separating data collection and preparation, feature engineering, probabilistic models, optimization and a query application; notebooks support exploration.

Stack: Python, pandas, NumPy, PyArrow, Parquet, OR-Tools, CBC, Jupyter, Requests

Public repository — Cartola Oracle

ClassPlanner — AI-assisted lesson planning

Web frontend, API, authentication and persistence connecting BNCC curriculum information and Gemini to lesson-planning workflows.

Application area: Education and lesson planning.

Users and context: Teachers.

Objective: Organize lesson plans using curriculum context, objectives, methodology and assessment.

Architecture: Frontend organized by pages and hooks; backend with controllers, services, repositories, routes, database and language model integration.

Stack: JavaScript, React, Node.js, Express, SQLite, Gemini, JWT, bcrypt, styled-components

Dente de Leão — interactive narrative prototype

Experimental browser 3D experience combining rendering, controls, camera, physics and narrative state.

Application area: Games, interactive narrative and 3D graphics.

Users and context: Interactive narrative experience exploration.

Objective: Build an exploration environment with interaction, physics and narrative elements.

Architecture: Input, world, character, camera, interaction and interface layers; visual assets and physics compose the 3D experience.

Stack: TypeScript, React, Vite, Three.js, React Three Fiber, Drei, Rapier, Zustand

Doctoral study — educational genetic algorithm

JavaScript exercise demonstrating population generation, fitness, selection, crossover and mutation against a target vector.

Application area: Evolutionary algorithm teaching and research.

Users and context: Academic experimentation.

Objective: Explore solution representation, evolution and evaluation.

Architecture: Genetic algorithm exercise exploring solution representation and evolution.

Stack: JavaScript

DVR Person Alert — vision and operational notifications

RTSP monitoring application with person detection, Telegram alerts, local controls, detection review and state/transport components.

Application area: Security, video monitoring and computer vision.

Users and context: Camera monitoring and event review.

Objective: Detect events, collect reviewed examples and adapt classification to the camera context.

Architecture: Video monitoring, management dashboard, temporal identity handling, event review and Telegram alert delivery in separate components.

Stack: Python, OpenCV, Ultralytics, YOLO, Flask, RTSP, FFmpeg, Telegram Bot API, psutil, PyTorch

Google Auth — integration configuration

Google authentication configuration used as integration support; no standalone application identified.

Application area: Authentication and Google service integration.

Users and context: Application integration setup.

Objective: Explore authentication and service access configuration.

Architecture: Authentication and integration configuration, with a study and access setup scope.

Stack:

Instabot — browser automation with persistent state

Local Instagram browser automation recording processed users and interactions, with configuration, persistent sessions and execution history.

Application area: Digital marketing and browser automation.

Users and context: Social media operational routines.

Objective: Automate interactions with persistent state and execution control.

Architecture: Configuration, session management, interaction collection and routine execution with persistent operational state.

Stack: JavaScript, Node.js, Playwright, Chrome DevTools Protocol, JSON, dotenv

Conversion CRM — conversations and assisted AI

Local CRM organizing campaigns, leads, conversation history and a SPIN-assisted flow with an output queue and Gemini integration.

Application area: Marketing, sales and customer relationships.

Users and context: Conversational sales and customer service operations.

Objective: Organize the sales funnel and support conversational decisions with history and suggested actions.

Architecture: API, database, services, SPIN decision engine and Gemini integration; event processing and a dashboard support the conversational workflow.

Stack: JavaScript, Node.js, Express, SQLite, Gemini, HTML, CSS

Launch With Me — product-validation marketplace

Next.js/Supabase platform modeling roles, campaigns and financial workflows for users completing product-validation missions and submitting evidence.

Application area: Entrepreneurship, marketplaces and product validation.

Users and context: Product creators and potential users.

Objective: Connect product proposals, discovery and validation in a web platform.

Architecture: Role-based routes, server operations, centralized integrations and Supabase persistence with functions and row-level access rules.

Stack: TypeScript, Next.js, React, Tailwind CSS, Supabase, PostgreSQL, Stripe, Zod, React Hook Form

Leaflet — web maps and routing

Map/routing experiment accompanied by a third-party Leaflet Routing Machine library; a reference for integration.

Application area: Maps, routing and geographic interfaces.

Users and context: Spatial information visualization and exploration.

Objective: Represent geographic data and routes in a web interface.

Architecture: Map interface and routing library integration, including studies and third-party examples.

Stack: JavaScript, HTML, Leaflet, Leaflet Routing Machine

Job Assistant — supervised automation and fit ranking

Local CLI collecting and ranking vacancies, organizing fit criteria and assisting applications with human review, typed configuration and SQLite history.

Application area: Careers, recruitment and supervised automation.

Users and context: Professional opportunity organization.

Objective: Collect opportunities and rank fit under user supervision.

Architecture: Browser automation, application workflow, fit ranking, configuration, database and supervised execution.

Stack: Python, Playwright, Pydantic, Typer, Rich, PyYAML, SQLite, pytest, Ruff

Logística — routing and geospatial visualization

Web prototype exploring points, farm polygons and routes using maps and genetic-algorithm experiments.

Application area: Logistics, routing and geospatial analysis.

Users and context: Travel planning and logistics operations.

Objective: Represent locations and routes on maps and support routing decisions.

Architecture: Map, drawing and routing components, supporting functions and genetic algorithm experiments.

Stack: JavaScript, React, Leaflet, Leaflet Routing Machine, Axios, Canvas

Personal finance — Open Finance and transaction analysis

Local Pluggy integration persisting bank transactions in SQLite and presenting expense, recurring-payment and daily-availability analyses with a demo dashboard.

Application area: Personal finance, Open Finance and transaction analysis.

Users and context: Personal financial organization.

Objective: Integrate and analyze transactions to support financial monitoring.

Architecture: Web server, financial data source integration, transaction analysis engine and dashboard.

Stack: Python, Python standard library, SQLite, Pluggy, JavaScript, HTML, CSS

Mocap — browser pose and movement estimation

Browser motion-capture experiments using video/webcam input, keypoints, skeletons, smoothing and 3D resources.

Application area: Human movement, pose capture and avatar interaction.

Users and context: Body interaction and movement visualization experiments.

Objective: Estimate pose in video and connect body landmarks to browser visualization.

Architecture: Inference and rendering coordination, video processing, pose estimation and movement visualization assets.

Stack: JavaScript, Vite, TensorFlow.js, MoveNet, BlazePose, MediaPipe, WebGL, p5.js, Three.js, VRM

Movimento 3D — model visualization

Exploratory browser scene and FBX-loading experiment for graphical resources and movement.

Application area: Computer graphics and model visualization.

Users and context: 3D scene and movement studies.

Objective: Explore 3D model presentation and movement.

Architecture: Visualization application with FBX model loading and 3D rendering.

Stack: JavaScript, HTML, Three.js, FBXLoader

TraderMind — quantitative research and model pipelines

Research laboratory and operational dashboard combining time-series collection, features, training, backtests, out-of-sample evaluation and API/MetaTrader5 integration.

Application area: AI, Machine Learning and MLOps for quantitative research.

Users and context: Financial-market model and strategy research and experimentation.

Objective: Structure the data, training, temporal validation, inference and model supervision lifecycle; integrate signals and policies with MetaTrader5.

Architecture: Studies and notebooks connected to an inference API, Python MetaTrader5 connector, configuration, supervision and operational dashboards.

Stack: Python, JavaScript, MetaTrader 5, pandas, NumPy, scikit-learn, XGBoost, Flask, SQLite, Jupyter, PyTorch, TensorFlow, LightGBM, CatBoost, SciPy, statsmodels, DEAP, joblib

OSORACLE — narrative knowledge organization

Obsidian vault organizing characters, chapters, references and dashboards for creative-project documentation.

Application area: Writing, narratives and knowledge organization.

Users and context: Writers and narrative world organization.

Objective: Structure information, entities and relationships in a narrative domain.

Architecture: Thematic organization of characters, chapters, bestiary, references and original narrative materials.

Stack: Obsidian, Markdown

Planejamento Xavier — planning product with AI

Typed frontend, database and backend functions connecting goals, activities, calendar, contextual AI and replanning.

Application area: Personal planning, productivity and professional development.

Users and context: People managing goals, activities and development plans.

Objective: Connect goals, capacity and execution through AI-assisted planning and review.

Architecture: Planning and context domains, interface components, APIs, workers, versioned persistence and simulation routines.

Stack: TypeScript, React, Vite, Tailwind CSS, Radix UI, TanStack Query, Supabase, PostgreSQL, Deno, Capacitor, Recharts, Zod

Live website — Planejamento Xavier

Planejamento Xavier — historical backup

Earlier React interface and operational scripts preserving a previous stage of product development.

Application area: Personal planning, productivity and professional development.

Users and context: People managing goals, activities and development plans.

Objective: Connect goals, capacity and execution through AI-assisted planning and review.

Architecture: Planning and context domains, interface components, APIs, workers, versioned persistence and simulation routines.

Stack: TypeScript, React, Supabase, Tailwind CSS

PoseNet / ml5 — ML reference collection

Local community examples for JavaScript ML concepts including pose, segmentation and generative models; study/reference material.

Application area: Machine Learning and pose estimation.

Users and context: Computer vision model studies and references.

Objective: Explore existing model integration to identify body landmarks.

Architecture: Examples organized by Machine Learning libraries and tasks; existing model integration.

Stack: JavaScript, ml5.js, p5.js, TensorFlow.js

React Native — starter environment

Generated starter application with the default screen and mobile development configuration.

Application area: Mobile development.

Users and context: Mobile interface and application studies.

Objective: Explore an initial React Native application structure.

Architecture: Initial mobile application template and development configuration.

Stack: TypeScript, React Native, React, Metro, Jest

Legacy scraper — initial offer collection

Earlier HTTP/HTML parsing scripts and tabular exports documenting the evolution toward a modular collection pipeline.

Application area: Data collection, offers and content distribution.

Users and context: Offer monitoring and publishing operations.

Objective: Structure offer information collection, processing and distribution.

Architecture: Collection core with site-specific adapters, execution, supervision, dashboard and persistence; legacy versions are separated.

Stack: JavaScript, Node.js, Axios, Cheerio, XLSX

Scrapers — offers and distribution pipeline

Node.js pipeline separating collection, site adapters, link processing, supervised distribution and persistence across SQLite and PostgreSQL/Supabase.

Application area: Data collection, offers and content distribution.

Users and context: Offer monitoring and publishing operations.

Objective: Structure offer information collection, processing and distribution.

Architecture: Collection core with site-specific adapters, execution, supervision, dashboard and persistence; legacy versions are separated.

Stack: JavaScript, Node.js, Axios, Cheerio, Puppeteer, Express, SQLite, PostgreSQL, Supabase, Telegram, WhatsApp Web

Skeleton from Video — pose experiments

Video-pose and skeleton-rendering experiments with BlazePose and keypoints, documenting the evolution toward Mocap.

Application area: Computer vision and movement analysis.

Users and context: Video and body representation experimentation.

Objective: Extract and visualize skeletons from video.

Architecture: Pose detection variants using BlazePose and representation of recorded body positions.

Stack: JavaScript, HTML, TensorFlow.js, BlazePose, p5.js

SpeedTest — exploratory network diagnostics

Node.js script integrating external speed tests and formatted ping/download/upload output.

Application area: Networks and connectivity diagnostics.

Users and context: Connection performance exploration.

Objective: Investigate connectivity measurements and their presentation.

Architecture: Exploratory execution and request routines for connectivity diagnostics.

Stack: JavaScript, Node.js, child_process, speedtest-net, fast-speedtest-api, Axios

Three.js — browser scenes and 3D models

Exploratory application for FBX loading, cameras, lighting and browser rendering, with Express serving interface assets.

Application area: Computer graphics and 3D web development.

Users and context: Scene, model and interaction studies.

Objective: Explore browser rendering of scenes and 3D assets.

Architecture: Scene, interaction, canvas and server for a 3D web experience.

Stack: JavaScript, Three.js, WebGL, FBXLoader, OrbitControls, Express

Unity — exploration folder

Folder present in the inventory without identifiable application code; it does not establish development experience on its own.

Application area: Game and interactive environment development.

Users and context: Development tool exploration.

Objective: Explore the Unity environment; implementation scope is limited in the inventory.

Architecture: Environment exploration; the inventory does not contain a detailed implementation to describe.

Stack:

Utilibox — tools and knowledge products

Shared web platform combining browser processing, external functions, persistence and commercial integrations for productivity, analysis, decisions, writing and knowledge organization.

Application area: Finance, employment, production engineering, decision making, education, writing, health, beauty, documents, design and marketing.

Users and context: General users, students, writers and professionals.

Objective: Turn calculations, methods and knowledge organization into accessible interactive tools.

Architecture: Domain-specific tools, Narra and UtiliNotas modules, shared pages and layouts, centralized policies and integrations and a Supabase backend.

Stack: JavaScript, TypeScript, Astro, Node.js, Supabase, PostgreSQL, Stripe, Gemini, CodeMirror, KaTeX, pdf-lib, Playwright

Open tool — utilinotas

Utilibox Manga — creative product variant

Utilibox variant reusing tools, writing and backend components to explore narrative and comics presentation.

Application area: Finance, employment, production engineering, decision making, education, writing, health, beauty, documents, design and marketing.

Users and context: General users, students, writers and professionals.

Objective: Turn calculations, methods and knowledge organization into accessible interactive tools.

Architecture: Domain-specific tools, Narra and UtiliNotas modules, shared pages and layouts, centralized policies and integrations and a Supabase backend.

Stack: JavaScript, TypeScript, Astro, Supabase, Stripe, CodeMirror, Gemini

Digital CV — professional presentation

JSON-driven CV and cover-letter interface with display/print modes and Vercel configuration. The empty PDF API does not establish server-side PDF conversion.

Application area: Professional presentation and web development.

Users and context: Recruiters and CV readers.

Objective: Present career history, skills and projects with language selection and PDF export.

Architecture: Main HTML with structured data, bilingual content, view controls and browser printing.

Stack: HTML, CSS, JavaScript, JSON, Vercel

Public repository — CV digital

Globalcom Certification OS — Agentic AI architecture for certification and compliance

Delivered solution architecture, currently being implemented, for educational certification and compliance, combining deterministic workflows, specialist agent orchestration, managed RAG, versioned evidence and integrations for school CEA + SEVP processes.

Application area: Education, certification, accreditation, compliance, document management, audit and Agentic AI.

Users and context: Globalcom managers, partners, schools and certification teams.

Objective: Design evidence-driven certification operations with specialist agents and human approval.

Architecture: Delivered architecture, being implemented, with deterministic workflows, 10 specialists, an orchestrator and an independent auditor; managed RAG, Supabase/PostgreSQL, persistent workers and API/MCP/RPA integrations.

Stack: TypeScript, React, Node.js, Python, FastAPI, Supabase, PostgreSQL, pgvector, RAG, MCP, Playwright, RLS, RBAC, Filas e workers

Agrodan / Agritech.NE — fruit counting and packing traceability platform

Worked as a platform developer during the Agrodan employment period, connecting fruit counting and tracking on the reject conveyor with ERP and sorting-machine data, an operational dashboard and Telegram alerts for batch traceability and changeover decisions. The final project report is dated March 2024.

Application area: Agribusiness, fruit production, packing, computer vision, traceability and operational efficiency.

Users and context: Sorting-machine operators, packing teams and Agrodan managers.

Objective: Integrate fruit counts and industrial data to monitor batches, indicate transitions and reduce interruptions.

Architecture: Camera-based counting and tracking application, ERP/sorting-machine API integration, weight estimation by size and variety, dashboard and Telegram notifications; dedicated hosting is included in the project scope.

Stack: Computer vision, Deep Learning, Cameras, APIs, ERP, Sorting-machine integration, Dashboard, Telegram, Dedicated server, CRISP-DM

Utilibox — applications and product domains

Utilibox is an independent project translating methods, calculations and tasks into digital tools. Its scope combines decision support, production engineering, finance, knowledge, creation and everyday utilities. The areas below present the central idea and representative application examples.

Multicriteria decisions and management · Production engineering, Lean and continuous improvement · Finance, investments and economic analysis · People, compensation and calculation rules · Knowledge, writing and learning · Visual creation, applied AI and marketing · Documents, files and digital productivity · Wellbeing, body metrics and pregnancy information · Beauty, formulation and consumption planning · Construction, resources and everyday planning · Time, organization and focus

Multicriteria decisions and management

Tools for structuring problems, comparing alternatives and prioritizing actions. The central idea is to make decision methods usable through interfaces with criteria, weights, rankings and visualizations, connecting quantitative analysis to management decisions.

Methods and topics: AHP, TOPSIS and PROMETHEE; Pareto, GUT, SWOT and expected monetary value.

Demonstrated capabilities: Criteria modeling, consistency, sensitivity and result presentation.

Examples: AHP Online · TOPSIS Online · PROMETHEE Online

Production engineering, Lean and continuous improvement

Applications for understanding workflows, identifying bottlenecks, reducing setup and tracking improvements. This group connects Lean tools to process metrics, capacity sizing and structured learning, relating to the ongoing Operational Excellence MBA.

Methods and topics: VSM, SMED, PDSA, A3, time study and queueing theory.

Demonstrated capabilities: Translating industrial methods into calculations, visual models and action tracking.

Examples: VSM Online · SMED Online · PDSA Manager

Related education: Operational Excellence MBA — in progress

Finance, investments and economic analysis

Simulators translating cash flows, rates, costs and periods into comparable indicators. Applications range from project appraisal and economic feasibility to financial planning, credit and scenario comparison.

Methods and topics: NPV, IRR, payback, WACC, break-even, interest and amortization.

Demonstrated capabilities: Financial mathematics, scenario modeling and explaining indicators to users.

Examples: Payback, NPV and IRR calculator · WACC calculator · Break-even simulator

People, compensation and calculation rules

Tools for exploring compensation, deductions, benefits and employment calculations. This domain demonstrates translating rules and parameters into simulators with checkable inputs and understandable results.

Methods and topics: Net salary, overtime, vacation pay, termination and tax and benefit estimates.

Demonstrated capabilities: Parameterized rules, conditional calculations, dates and information organization.

Examples: Brazilian net salary calculator · Overtime calculator · FGTS termination balance and 40% penalty

Knowledge, writing and learning

Products organizing ideas, content and relationships while retaining context throughout writing and study. Narra and UtiliNotas support knowledge organization; flashcards add review planning and spaced repetition.

Methods and topics: Entities and relationships, Markdown, connected notes, SM-2 and content export.

Demonstrated capabilities: Domain modeling, local persistence, editing and learning experiences.

Examples: Narra · UtiliNotas — connected Markdown editor · Anki flashcards — spaced repetition

Visual creation, applied AI and marketing

Tools connecting content creation, visual editing and commercial analysis. This experience combines browser graphics interaction, AI content generation and indicators for understanding customer acquisition and economic value.

Methods and topics: Vector and 3D editing, AI-generated carousels, LTV and CAC.

Demonstrated capabilities: Creative interfaces, AI integration and translating growth metrics into tools.

Examples: Online 3D editor · AI Instagram carousel generator · Customer lifetime value and acquisition cost — Google Ads

Documents, files and digital productivity

Utilities simplifying recurring tasks with documents, images and formats. The central idea is to reduce workflow steps through direct interfaces for manipulating files and producing reusable outputs.

Methods and topics: Conversion, compression, PDF splitting and merging, images and QR-code generation.

Demonstrated capabilities: File processing, formats, export and interaction workflow design.

Examples: Merge PDFs · JPG/PNG image compressor · QR code generator

Wellbeing, body metrics and pregnancy information

Informational calculators for exploring body metrics, physical activity, cycles and pregnancy timelines. This group extends development experience into domains with units, reference ranges and temporal relationships.

Methods and topics: BMI, energy expenditure, running pace, gestational age and weekly timelines.

Demonstrated capabilities: Parameterized calculations, unit conversion and presentation of informational estimates.

Examples: BMI calculator · Running pace calculator · Pregnancy duration calculator

Beauty, formulation and consumption planning

Simulators focused on proportions, consumption and beauty-service planning. This domain demonstrates calculation logic and parameter modeling applied to practical service and material-use situations.

Methods and topics: Mixing ratios, product quantities, colorimetry and hair-care schedules.

Demonstrated capabilities: Parameter modeling, consumption estimates and context-oriented interfaces.

Examples: Dye and developer mixing calculator · Hair dye quantity calculator · Hair-care schedule calculator

Construction, resources and everyday planning

Applications estimating materials, costs, consumption and quantities in household tasks and everyday planning. The engineering connects units, waste allowances, measurements and prices to outputs helping users plan resources.

Methods and topics: Area, price per square meter, materials, electricity use and quantity sizing.

Demonstrated capabilities: Quantitative estimation, conversions, consumption rules and cost comparison.

Examples: Floor and tile calculator · Paint quantity calculator · Electricity consumption calculator

Time, organization and focus

Tools organizing duration, dates, deadlines and work cycles. This group translates time operations and focus methods into simple interactions for personal planning and productivity.

Methods and topics: Business days, intervals, stopwatch, countdown and Pomodoro.

Demonstrated capabilities: Temporal logic, interface state, events and productivity routine design.

Examples: Online Pomodoro · Business-day calculator · Stopwatch and countdown timer

Shared product engineering

Product engineering combines Astro, JavaScript and TypeScript with interactive interfaces, data modeling and calculation logic. Modules requiring these capabilities use IndexedDB local persistence, Supabase/PostgreSQL services, Gemini AI integrations, content editing and document generation. This connects domain knowledge to software development and user experience.

See the highlighted Utilibox cases

Capabilities and their project origins

Capabilities organized by specialty, with applications, methods and source projects. Each group connects technical work to its professional, academic or independent context; links open the corresponding cases.

Agentic AI, multiagent systems and governance · LLMs, AI applications and conversational automation · Machine Learning, quantitative research and experimentation · MLOps, pipelines and model lifecycle · Computer vision and industrial integration · Data engineering, ETL and ERP integration · Web and mobile software engineering and products · Agriculture, operational management and enterprise systems · Multicriteria decisions, optimization and operations research · Lean, continuous improvement and economic analysis · Knowledge management, learning and domain modeling · Infrastructure, security and technology leadership · 3D interfaces, maps and interactive experiences

Agentic AI, multiagent systems and governance

Solution architecture with specialized roles, agent coordination, shared context and decision controls. Planejamento Xavier provides rule-based coordination; Globalcom extends this experience through a delivered certification and compliance architecture, currently being implemented.

Applied capabilities

- Responsibility decomposition, orchestration and conflict resolution
- RAG, document retrieval, MCP and system integration in the Globalcom design
- Persistent workflows, independent audit, permissions and human approval

Source cases and contexts

- Six-agent planning council — Planejamento Xavier

  Independent projects, prototypes and research

  Implemented a council of six specialized agents to support planning and replanning, with evidence for each recommendation, deterministic conflict resolution and traceable actions.
- Globalcom · Agentic AI architecture and specialist orchestration
- Globalcom · Managed RAG and document intelligence
- Globalcom · Workflow, events and persistent execution
- Globalcom · AI governance, security and continuous audit
- Globalcom · API, MCP and RPA integrations for external operations
- Globalcom · Certification domain and product modeling
- Globalcom · Consulting, solution architecture and roadmap

Technologies and methods: TypeScript · Deno · Supabase · typed contracts · conflict resolution · React · Node.js · Python · FastAPI · PostgreSQL · pgvector · RAG · MCP · Playwright · RLS · RBAC · Queues and workers

LLMs, AI applications and conversational automation

Language-model integration in planning, education and customer relationship products. This work connects AI responses to application domains through data structures, validation and failure handling, turning model output into usable actions.

Applied capabilities

- Provider integration, model selection and fallback
- Structured outputs, operation contracts and reference validation
- Conversation context and AI-assisted plan review

Source cases and contexts

- Shared AI integration engine — Planejamento Xavier

  Independent projects, prototypes and research

  Developed a shared AI integration layer with provider adapters, structured responses, validation, time budgets and task-specific failure handling.
- AI-assisted plan review and operation contracts

  Independent projects, prototypes and research

  Modeled AI-assisted review proposals with reference validation, alternatives, operations and follow-up, connecting suggested changes to the actual plan.
- AI-assisted lesson planning — ClassPlanner

  Independent projects, prototypes and research

  Integrated Gemini lesson-plan generation into a web application using pedagogical context, Brazilian BNCC curriculum references and structured objectives, methodology and assessment.
- Hybrid conversational CRM — Instagram Conversion CRM

Independent projects, prototypes and research

Built a hybrid AI-and-rules conversation flow with interaction history, action contracts, failure handling and human handoff.

Technologies and methods: TypeScript · Deno · LLM APIs · JSON · telemetry · fallback · Supabase · operation contracts · reference validation · React · Node.js · Express · SQLite · Gemini · BNCC · conversation history

Machine Learning, quantitative research and experimentation

Applied research combining data preparation, feature selection, evolutionary algorithms, predictive models and experimental evaluation. EFS-AHP belongs to doctoral research; TraderMind explores time series and signal ensembles in quantitative research, without implying verified financial returns.

Applied capabilities

- Feature selection, AHP, baselines and model comparison
- Evolutionary algorithms, ensembles and signal combination
- Feature engineering, temporal evaluation and data leakage prevention

Source cases and contexts

- EFS-AHP — doctoral research framework

  Doctoral research

  Developed EFS-AHP as doctoral research combining feature selection, evolutionary algorithms and multicriteria decision analysis, comparing ML alternatives and analyzing performance, dimensionality and computational cost.

- TraderMind — experiments and inference integration

  Independent projects, prototypes and research

  Structured quantitative experiments and inference components, connecting data preparation and model/policy evaluation to an API and the MetaTrader5 terminal.

- Financial signal ensemble — TraderMind

  Independent projects, prototypes and research

  Designed multiagent policy experiments with signal voting, quorum and simulated annealing, separating calibration, temporal validation and out-of-sample evaluation.

Technologies and methods: Python · scikit-learn · pandas · NumPy · SciPy · DEAP · AHP · XGBoost · Flask · MetaTrader5 · simulated annealing · temporal validation

MLOps, pipelines and model lifecycle

TraderMind centers on the engineering connecting data, training, validation, artifacts, inference and model monitoring. MetaTrader5 integration is one component of this workflow; the cases describe research and demo operation, separating experimentation, API and execution responsibilities.

Applied capabilities

- Data contracts, feature preparation and out-of-sample validation
- Checkpoints, artifact persistence and model serving
- Process supervision, retraining, updates and observability

Source cases and contexts

- Data pipelines, features and quality contracts

  TraderMind · independent MLOps and quantitative research project

  Integrated extraction, data updates, dataset construction and feature selection into operational TraderMind pipeline stages.

- Training, checkpoints and artifact management

  TraderMind · independent MLOps and quantitative research project

  Organized experiments, training and execution recovery while preserving the relationship between models, features and parameters.

- Temporal validation, leakage and out-of-sample evaluation

  TraderMind · independent MLOps and quantitative research project

  Separated selection, calibration and temporal evaluation to compare models and policies without contaminating the final evaluation period.

- Model serving and MetaTrader5 integration

  TraderMind · independent MLOps and quantitative research project

  Connected ML artifacts, an inference service and the desktop terminal to bridge research and demonstration operation.

- Orchestration, retraining and operational updates

TraderMind · independent MLOps and quantitative research project

Coordinated data, training, evaluation and serving cycles through distinct execution modes.

- Observability and model follow-up

  TraderMind · independent MLOps and quantitative research project

  Connected the operational dashboard, pipeline stage health and performance review using recent data.

- MetaTrader5 desktop connector — TraderMind

  Independent projects, prototypes and research

  Built a Python connector to the MetaTrader5 desktop terminal, integrating market/account data with the documented research and execution workflow.

Technologies and methods: Python · pandas · NumPy · Parquet · DEAP · EFS-AHP · scikit-learn · XGBoost · LightGBM · CatBoost · joblib · checkpoints · temporal validation · walk-forward · OOS · backtesting · Flask · REST APIs · MetaTrader5 · supervisor · scheduling · logs · pipeline automation · dashboard · monitoring · SQLite · local APIs

Computer vision and industrial integration

Vision applications range from video detection and notifications to browser pose estimation and industrial fruit counting. At Agrodan/Agritech.NE, the developer contribution connects the platform to ERP, sorting-machine, batch and operational indicator data; industrial outcomes are those reported for the project.

Applied capabilities

- Video inference, feedback datasets and classifier training
- Pose estimation, client-side processing and visualization
- Counting, traceability, volume reconciliation and industrial alerts

Source cases and contexts

- DVR — feedback, datasets and classifier training

  Independent projects, prototypes and research

  Implemented a computer-vision workflow with detection, example review and custom classifier training from a pretrained model, integrating epoch monitoring, checkpoints, temporal stabilization and persistent alerts.

- Mocap — browser inference and visualization

  Independent projects, prototypes and research

  Developed browser pose-estimation prototypes integrating inference, keypoint visualization and temporal smoothing to explore motion capture and 3D interaction.

- Computer vision platform development for packing operations

  Agrodan · Agritech.NE applied R&D project · platform developer

  Worked as a developer on the platform project during my Agrodan employment period, connecting fruit counting and tracking to operational packing monitoring.

- ERP, sorting-machine integration and batch traceability

  Agrodan · Agritech.NE applied R&D project · platform developer

  The platform connects incoming batches, packed volume and rejected fruit to track fruit origin and support batch transitions.

- Dashboard, alerts and operational decision support

  Agrodan · Agritech.NE applied R&D project · platform developer

  Developed the monitoring and notification platform to provide operators with processing information and batch-change timing.

- Project outcomes and industrial process improvement

  Agrodan · Agritech.NE applied R&D project · platform developer

  Platform integration into the packing workflow produced project-level outcomes recorded in the final report, covering throughput, downtime and operating costs.

Technologies and methods: Python · OpenCV · Ultralytics/YOLO · PyTorch · Flask · RTSP · JavaScript · TensorFlow.js · MoveNet · BlazePose · MediaPipe · WebGL · Computer vision · Deep Learning · Cameras · APIs · ERP · Sorting-machine integration · Dashboard · Telegram · Dedicated server · CRISP-DM

Data engineering, ETL and ERP integration

Operational data integration supporting corporate indicators and workflows at Agrodan. The engineering covers batch imports, persistence, APIs and update routines, connecting ERP records to web applications, analytics and management decisions.

Applied capabilities

- Data modeling and SQL queries

- Modular ETL, batch ingestion and data quality
- ERP integration, error traceability and operational updates

Source cases and contexts

- Modular ETL and operational data quality

  Professional experience — Agrodan

  Structured integration pipelines with reusable importers, batch processing and domain rules for production, packing, quality and people, organizing operational data for analysis and decisions.

Technologies and methods: Node.js · SQL · Oracle · Sequelize · ETL · batch import

Web and mobile software engineering and products

Development of systems with domain rules, interfaces, APIs and persistence across corporate, hospital, academic and entrepreneurial contexts. SIAgro extends this experience to mobile collection; independent products add entity modeling, authentication, services and commercial integrations.

Applied capabilities

- Web applications, API architecture and relational persistence
- React Native, navigation, operational collection and local storage
- Java EE systems, indicator management, gamification and digital products

Source cases and contexts

- SIAgro — mobile operational data collection

  Professional experience — Agrodan

  Developed SIAgro mobile modules for work orders, production records and quality using React Native/Expo, business-object modeling, local persistence and operational indicators.

- Hospital indicators and academic gamification

  Hospital internship and university project

  During the hospital internship, developed a Java EE web system for indicators and assessments. At university, built a project-management application with gamification and persistence, exploring progression rules and task organization.

- Marketplace, automation and 3D interaction

  Independent projects, prototypes and research

  Structured products and prototypes with access rules, integrations and supervised workflows, including a marketplace, commercial collection, personal finance and a narrative 3D experience.

Technologies and methods: React Native · Expo · AsyncStorage · React Navigation · APIs · Java EE · EJB · JSF · PrimeFaces · JPA/Hibernate · JPQL · Quartz · Next.js · Supabase · PostgreSQL · Stripe · Playwright · SQLite

Agriculture, operational management and enterprise systems

Within SIA/SIAgro, technology supports agricultural planning, irrigation, spraying, purchasing, people, quality and packing. The cases show business rules translated into modules, data and interfaces, connecting software development to professional digital transformation and management experience at Agrodan.

Applied capabilities

- Agronomic rules, irrigation modeling and harvest forecasting
- Offline collection, synchronization, field audits and geospatial data
- Purchasing, assessments, business units, commercial indicators and quality management

Source cases and contexts

- PLANAGRO — inputs, applications and agronomic rules

  Professional experience — Agrodan

  Developed agricultural planning components in PLANAGRO, translating input application, dosage, withholding periods, drift and budgets into rules and interfaces connected to operational data.

- Irrigation — scheduling, genetic algorithms and Kc modeling

  Professional experience — Agrodan

  Explored irrigation scheduling, genetic algorithms and neural modeling of crop coefficient Kc, working with variable encoding, normalization, network parameters and operational backend integration.

- Spraying audit — maps and field execution

  Professional experience — Agrodan

  Built spraying-audit components using spatial and operational data, relating trajectories, speed, field blocks and services to support investigation of field execution.

- Dry matter — harvest forecasting and readiness

  Professional experience — Agrodan

  Implemented dry-matter analysis and harvest-forecast components connecting readings, varieties and parameters to charts and monitoring rules.

- Field notebook and purchasing — local data and synchronization

  Professional experience — Agrodan

  Developed field-data collection and purchase-request workflows with local storage, process components and synchronization routines integrated with domain-specific backend services.

- UGB management, people and improvement suggestions

  Professional experience — Agrodan

  Modeled business-unit and people-management applications and improvement-suggestion workflows combining organizational records, history, assessments, roles and documents.

- Commercial monitoring, forecasts and pallet movements

  Professional experience — Agrodan

  Developed commercial and logistics components connecting records and forecasts to management interfaces and operational pallet/container data.

- QMS — audits, records and quality management

  Professional experience — Agrodan

  Developed quality-management modules organizing audits, records and roles with process follow-up and local storage.

- Field quality — checklists and operational assessments

  Professional experience — Agrodan

  Developed field-quality assessment workflows with checklists, responses and integration between interfaces and the domain backend.

- Packing and box handling — operational interfaces

  Professional experience — Agrodan

  Developed packing and box-handling interfaces organizing access to operations and connecting events to monitoring components.

Technologies and methods: React · JavaScript · Node.js · operational APIs · agronomic rules · genetic algorithms · neural network · normalization · geospatial data · trajectory · speed · curves and equations · charts · field data · Dexie · IndexedDB · synchronization · organizational modeling · assessments · history · commercial data · pallets · containers · domain APIs · socket events · operational interfaces

Multicriteria decisions, optimization and operations research

Analytical methods translated into tools for comparing alternatives, sizing capacity and making constrained decisions. Utilibox provides AHP, TOPSIS, PROMETHEE and queueing applications; Cartola and logistics extend this experience to quantitative decisions and applied optimization.

Applied capabilities

- Weights, consistency, rankings and sensitivity analysis
- Queueing theory, capacity and service sizing
- Constrained optimization, probabilistic analysis and routing

Source cases and contexts

- Utilibox — multicriteria analysis, queues and applied methods

  Independent projects, prototypes and research

  Implemented decision-support and learning tools using multicriteria methods, queueing theory, simulation, financial mathematics and spaced repetition, translating models into explainable interfaces and calculations.

- Cartola and logistics — decisions under constraints

  Independent projects, prototypes and research

  Developed pipelines connecting data and optimization to decisions, exploring team selection under budget/formation constraints and routing with geographical visualization.

- AHP Online — multicriteria decisions and consistency

  Independent product — Utilibox

  Developed a multicriteria decision tool organizing criteria, comparisons and alternatives into an analysis and project-export workflow.

- TOPSIS and PROMETHEE — rankings and sensitivity

Independent product — Utilibox

Implemented tools for comparing alternatives with configurable criteria, weights and preferences, applying multicriteria methods to management decisions.

- Queueing theory — capacity, waiting and service sizing

  Independent product — Utilibox

  Developed calculators and visualizations relating demand, service capacity and waiting, supporting study and comparison of service configurations.

Technologies and methods: Astro · JavaScript · TypeScript · AHP · TOPSIS · PROMETHEE · simulation · Python · pandas · Parquet · OR-Tools/CBC · React · Leaflet · comparison matrices · consistency · sensitivity analysis · operations research · queueing models · sensitivity curves

Lean, continuous improvement and economic analysis

Production engineering and management methods implemented in Utilibox digital tools. This experience connects value streams, setup reduction, structured problem solving, prioritization and economic evaluation; the Lean tools also relate to the ongoing Operational Excellence MBA.

Applied capabilities

- VSM, SMED, PDSA and A3 for process analysis and improvement
- Pareto, GUT and SWOT for prioritization and diagnosis
- NPV, IRR, payback and scenarios for investment evaluation

Source cases and contexts

- VSM, SMED, PDSA and A3 — production engineering and continuous improvement

  Independent product — Utilibox

  Translated process-improvement methods into digital tools for mapping flows, analyzing setup, structuring problems and tracking learning cycles.

- Economic analysis — NPV, IRR, payback and scenarios

  Independent product — Utilibox

  Implemented economic-analysis tools transforming cash flows, costs and financing conditions into comparable indicators.

- Pareto, GUT and SWOT — prioritization and management analysis

  Independent product — Utilibox

  Developed management tools organizing problems, criteria and scenarios into rankings, charts and decision structures.

Technologies and methods: Astro · JavaScript · Lean · VSM · SMED · PDSA · A3 · financial mathematics · cash flows · investment analysis · Pareto · GUT · SWOT · EMV · decision visualization

Knowledge management, learning and domain modeling

Products organizing knowledge, relationships and learning routines. Narra, UtiliNotas and flashcards demonstrate entity modeling, context and persistence alongside study methods, spaced review and content export.

Applied capabilities

- Entities, relationships, context and knowledge organization
- Content persistence, Markdown and export
- Spaced repetition, review planning and study tracking

Source cases and contexts

- Narra and UtiliNotas — domain and knowledge applications

  Independent projects, prototypes and research

  Built narrative and knowledge-organization modules with entity relationships, contextual assistance, response traceability and note interoperability.

- Narra and UtiliNotas — knowledge, context and persistence

  Independent product — Utilibox

  Built narrative and knowledge-organization applications with related entities, linked notes and contextual assistance.

- Flashcards — spaced repetition and study tracking

  Independent product — Utilibox

  Implemented a study application organizing cards and reviews through spaced repetition, retention tracking and export.

Technologies and methods: JavaScript · TypeScript · Astro · entities and relationships · Markdown · IndexedDB · entity modeling · contextual AI · SM-2 · local persistence · TSV · PDF

Infrastructure, security and technology leadership

ICT management experience at Agrodan covers enterprise environments, support, servers, networks and operational continuity. This experience complements technology leadership, connecting operations, technical teams, enterprise systems and business needs.

Applied capabilities

- Windows/Linux, Active Directory, networks and server administration
- Monitoring, service desk, backup, recovery and availability
- Team management, prioritization and technology–operations integration

Source cases and contexts

- IT operations management — Agrodan

  Professional experience — Agrodan

  Managed IT operations, integrating support, servers, networks, continuity, ERP and IoT initiatives with delivery of technology services and solutions.

Technologies and methods: Windows Server · Linux · Active Directory · Zabbix · GLPI · networks · IoT

3D interfaces, maps and interactive experiences

Exploration of spatial interfaces, maps, models and interactive worlds in independent projects. This work combines browser rendering, interaction, physics and geographic visualization; the cases retain their prototype and study context.

Applied capabilities

- 3D scenes, interaction components and application state
- Models, animation, physics and WebGL rendering
- Maps, routes and geospatial visualization

Source cases and contexts

- Dente de Leão — interactive narrative world

  Independent projects, prototypes and research

  Built a narrative 3D prototype with character controls, camera, interaction, physics and HUD, organizing state and rendering with React Three Fiber, Rapier and Zustand.

- Maps, models and spatial exploration

  Independent projects, prototypes and research

  Explored spatial visualization and 3D models with Leaflet and Three.js, building map, drawing, routing and scene-interaction interfaces.

Technologies and methods: TypeScript · React Three Fiber · Three.js · Rapier · Zustand · JavaScript · React · Leaflet · Canvas · FBX

Algorithms, data structures and problem solving

120 hours of complementary LeetCode practice covering arrays, hashing, binary search, trees, graphs, heaps, dynamic programming and backtracking. This experience supports complexity analysis and the selection of data structures and solution strategies.

See complementary experience and reference challenges

Project and career context

The projects cover professional applications at Agrodan, doctoral EFS-AHP research, hospital management during the internship and university project gamification. Independent initiatives include digital products, decision tools, education, finance, automation and interactive experiences.